

Benchmark run full detail

Run ID: 20260429T002305Z_pure_full

Generation model: ollama / qwen2.5:7b-instruct

Standard validator: gemini / gemini-2.5-flash

Layer 1 uses sentence-transformer greedy one-to-one matching at threshold 0.55. Layer 2 uses a checklist-based Gemini judge with strict and weighted scoring.

Document: 0000_cctns

Part A - Document header

Document ID	0000_cctns
Generation model	ollama / qwen2.5:7b-instruct
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	100
Generated requirement count	71
Matched count	41
Clarification rounds	requested 2 / used 2
Clarification chunks	2
Semantic P / R / F1	0.5775 / 0.4100 / 0.4795
LLM strict P / R / F1	0.2535 / 0.1800 / 0.2105
LLM weighted P / R / F1	0.4930 / 0.3500 / 0.4094
Evidence / Propositions / Gap	128 / 71 / 0
Grounded after rewrite	71

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 0000 - cctns system, and the goal is to capture the main requirements and constraints for document 0000_cctns.

[SYSTEM] Can you provide examples of frequently-executed transactions in the system to help us understand which processes should have fewer interactions?

[USER] Our users frequently need to place orders and check their order status. Ideally it would be great if these processes could be completed in just a few clicks. For placing an order, we should streamline the process so that customers can add items to their cart, proceed to checkout with minimal steps, and complete payment without too many interactions. Similarly, checking order status should only require logging in or entering an order number, then viewing the current status page which displays all relevant information like delivery date and tracking details. We've had issues where users get frustrated if they have to go through multiple screens just to find out their order status or complete a purchase.

[SYSTEM] Are there any specific steps or information that users need to provide during the order placement process that could be simplified or pre-filled based on their previous interactions?

[USER] We've had issues with customers having to re-enter the same information every time they place an order. It would be great if we could pre-fill some of this data based on their previous interactions. For example, when a customer logs in and starts a new order, it should recognize them and auto-populate fields like shipping address and payment method where applicable. This will save them time and reduce errors. Ideally, it would also allow us to show any saved payment details securely so they can just confirm or change if needed.

[SYSTEM] Are there any common scenarios where users might need to place multiple orders simultaneously or manage a list of items across different orders?

[USER] We need the system to handle scenarios where users might place multiple orders simultaneously. For example, a user shopping on their mobile device could add several items to their cart and checkout all at once. It should be able to manage a list of items across different orders, so that if they start one order but then want to make another, they can easily switch between them without losing any items. This is important because our users expect a seamless shopping experience, especially when they are in a hurry or buying multiple items at once.

[SYSTEM] How important is it to allow users to customize the menu contents based on their role or preferences within the system?

[USER] It's really important that users can customize their menu contents based on their role or preferences. For example, a manager might want to see different options than an entry-level user. We need to ensure the page length is appropriate for each task, and we should use color carefully, considering accessibility issues. Familiar terms are key — navigation links should be intuitive, using language that users will recognize from their experience with similar systems.

[SYSTEM] Could you provide more details on the types of non-text media objects that are commonly used in the system, so we can ensure they are all appropriately described with text equivalents?

[USER] We need to ensure that all non-text media objects are appropriately described with text equivalents. For example, graphical images or videos should have alternative descriptions provided. This is important for accessibility and ensuring everyone can understand the content. We also want to make sure our user interfaces allow activation of controls through a variety of input devices, not just keyboards or mice. Additionally, we need to follow ISO 9241-303 guidelines for screen text legibility to ensure readability.

[SYSTEM] Given the importance of accessibility, could you provide more details on how users with visual impairments typically interact with your system currently, and what specific features or tools they might rely on?

[USER] We know that users with visual impairments rely heavily on screen readers and other assistive technologies. Currently, some of our pages have a splash screen which can be an issue because it might not always allow for easy navigation using these tools. Ideally, we should offer a skip link to bypass the splash screen so they can get straight to the content they need. Additionally, placing page titles consistently in a prominent location would help them understand where they are and what information is available. We also want to avoid horizontal scrolling as much as possible because it can be difficult for users with visual impairments to navigate using screen readers. Lastly, we should ensure that navigation components like menus and links are placed consistently across the application so that users can predict where to find important features.

[SYSTEM] Can you provide examples of common user actions where resizing text would be particularly helpful, and how it might enhance their experience with the application?

[USER] Sure! Resizing text would be particularly helpful for users who have visual impairments or just prefer a larger font size. For example, when reading long articles or filling out forms, being able to increase the text size can make it much easier on the eyes and improve readability. This is especially important in our Help Module where detailed instructions are provided; making sure these instructions are easily readable could significantly enhance user experience. Additionally, using frames with care is crucial. We should ensure that when users navigate through different sections of the application, they don't encounter issues like broken back button functionality or bookmarking problems. It's also important to make it clear when individualization and adaptation features are in use, so users know their preferences have been applied correctly.

[SYSTEM] Could you describe the typical workflow of a user who frequently needs to skim through long text pages, and how our current implementation might be hindering their ability to quickly find relevant information?

[USER] Our users often need to quickly scan through long text pages for specific information. Currently, the system can be a bit cumbersome because it has too many links that make the text hard to read at a glance. We should provide an alternative text-only page as well, which would allow them to skim without getting distracted by hyperlinks. To support skimming better, we need clear links, bulleted lists, highlighted keywords, and logical headings. The links themselves should be marked with dashed underlines instead of solid ones to avoid overwhelming the user. It's important that our interface follows a consistent design so users can easily navigate and find what they're looking for without feeling lost or confused.

[SYSTEM] How can we ensure that the context-sensitive help material is easily accessible and relevant to users when they need it most during their tasks in the application?

[USER] We need to ensure that users can easily find help when they need it. For instance, having a site map would be great for giving an overview of the application structure. Each page should have a link back to the home or a recognizable landmark page. The context-sensitive help material is crucial — we want detailed guidance for every action and scenario in the app. To make sure users can understand the text quickly, we'll use short sentences, break up long texts into chunks, and present information as bullet points.

[SYSTEM] Can you provide examples of specific tasks or scenarios where users might need context-sensitive help, and what kind of assistance would be most beneficial for them?

[USER] We need to ensure that users can easily understand where they are in the application. For example, when a user is filling out a form for creating an account, it should clearly indicate which step they're on and how many steps remain. It's important that the system provides context-sensitive help at these points, like tooltips or inline instructions, to guide them through the process without confusion. Additionally, we need to make sure that our text quality is top-notch; spelling and grammar must be impeccable so as not to impede readability. This is especially crucial for users who might have special needs, such as those with visual impairments or cognitive disabilities. For instance, screen readers should be able to navigate through the application smoothly without any issues. We've had issues in the past where poorly written text caused confusion and slowed down user interactions. Ideally, it would be great if we could offer a variety of support features both within the app and through a browser interface. If someone needs help with something specific, they should be able to access that information quickly without having to navigate away from their current task.

[SYSTEM] Could you provide more details on how the application currently handles navigation for users with cognitive disabilities, and what improvements could be made to better support this user group?

[USER] We need to ensure that our application follows the principles of human perception when designing pages. It's important for users with cognitive disabilities to have a clear understanding of where they are in the navigation structure. Each page should provide this information clearly and succinctly. Ideally, it would be great if we could minimize the number of steps needed to reach any piece of content while still respecting different mental models and tasks that our users might have. Currently, we've had issues with too many clicks required to get to certain sections, which can be frustrating for everyone but especially problematic for those with cognitive disabilities. We should aim to streamline navigation as much as possible. Additionally, it's crucial that the application uses a consistent set of user interface rules throughout. This will help create a familiar and common look and feel, making it easier for all users to navigate.

[SYSTEM] Could you provide examples of how the application currently distinguishes within-page links from external ones to ensure they are clearly marked and accessible for users with visual impairments?

[USER] Currently, within-page links are underlined and appear in a different color than external links to make them stand out. We use color carefully, considering that some users might have difficulty distinguishing certain colors, so we rely on both color and text to convey information. For example, when a link opens a new window or tab, it includes an icon next to the text to indicate this, ensuring that all users can easily identify such links.

[SYSTEM] Could you provide examples of specific scenarios where users might attempt unauthorized access to cases, and how logging these attempts could help in security audits?

[USER] In our system, there are several scenarios where unauthorized access to cases might occur. For example, a user with limited permissions might try to view or modify a case they do not have access to. Another scenario could be an external attacker attempting to gain access through weak passwords or exploiting vulnerabilities in the application. Logging these attempts is crucial because it helps us track who tried what and when, which is vital for security audits. By exporting audit trails for specific cases without affecting the stored data, we can review these logs later if needed. This ensures that all activities related to cases are recorded, making our system legally admissible and aiding in data recovery efforts. The audit trail must be unalterable—no one should be able to change or delete it, ensuring its integrity.

[SYSTEM] Could you describe the process for creating and managing user profiles in our application, and how this might change based on different roles or departments within the organization?

[USER] For user profiles in our application, we need a way for users to log any defects or enhancement requests and track them. This is important because it helps us address issues quickly and improve the product based on customer feedback. Regarding profile management, only super-users should be able to change security attributes like access rights, security levels, privileges, and password management. Users should have the ability to see, modify, and delete their own profiles as needed. This ensures that each user can manage their own information while maintaining a high level of security through strict control over who can make changes.

[SYSTEM] Can you provide examples of how the current system handles data validation for user input, and what specific challenges or pain points have been encountered during this process?

[USER] Currently, the system handles data validation by checking user input against predefined rules. For example, we allow users to define their own default values for certain fields, which is really helpful when they need to enter similar information repeatedly. However, there have been some challenges with this approach. Sometimes, error messages can be unclear or too technical, leading to confusion among our users. We've had issues where the system doesn't provide clear instructions on how to correct the input, making it difficult for them to resolve errors themselves. Ideally, we would like the system to give more actionable feedback and suggest possible fixes. Additionally, while persistent defaults are useful, there have been times when they caused problems because users forgot to update their settings. We need a way to ensure that these defaults can be easily changed or overridden if necessary. Overall, registration complaints often come up due to misunderstandings about the validation rules, so we're looking for ways to improve both the clarity of our error messages and the overall user experience.

[SYSTEM] Could you provide more details on how the system currently handles user-defined default values and any issues that arise from this approach during data validation?

[USER] Currently, the system allows users to define default values which are quite useful for streamlining processes. However, we've had issues with data validation when these defaults don't align with our international standards. For example, AUTO-0082 requires us to consult specific International Standards for guidance, but sometimes user-defined defaults can conflict with those guidelines. Ideally, it would be better if the system could automatically check against these standards during validation. Registration citizens often use this feature to report complaints to the police, and while it's helpful, we need a way to ensure that any default values entered by users are compliant with our legal requirements for evidence collection and investigation. AUTO-0001 emphasizes the importance of following these procedures, so we must find a balance between user convenience and strict adherence to regulations. Additionally, we've noticed that relying solely on color coding can be problematic, as it should never be the only means of conveying important information. That's why we need robust validation checks in place to catch any discrepancies.

[SYSTEM] How can we improve the clarity and specificity of error messages when user-defined defaults conflict with international standards during data validation?

[USER] We need clear and specific error messages when user-defined defaults conflict with international standards. For example, if a user sets a date format that doesn't match the ISO standard, we should inform them about the correct format and suggest using something like YYYY-MM-DD. Ideally it would highlight the incorrect part of their input directly in the field. That's important to us because users might not be familiar with international standards and need guidance on how to adjust their settings.

[SYSTEM] Could you provide more details on the types of specialist software that will be integrated with our system and how their interface guidelines should influence the design?

[USER] We need the system to be highly customizable and user-configurable where possible. This is important because we might integrate with various specialist software tools that have specific interface guidelines. For example, our users should be able to adjust settings according to their needs, making sure everything remains intuitive and easy to use for both end-users and administrators. The design must follow broad guidelines such as supporting multilingual interfaces, which will cater to a diverse user base. Additionally, the system should ensure that no operational data is lost in case of equipment or communication network failures.

[SYSTEM] Could you provide more details on the expected range of technical contexts for opening and downloading times, such as different devices or network conditions, to help us understand better how to meet these performance capacity requirements?

[USER] We need the system to handle different devices and network conditions smoothly. For cases accessed within the last two months, it should be able to retrieve and display them within 5-8 seconds — this is crucial for a good user experience. Even when there are many cases or limited storage capacity, we expect this performance. For older cases not accessed in the past two months, retrieval and display should happen within 20 seconds. This ensures that while less frequently used data can take longer to load, it still meets our needs without causing significant delays.

[SYSTEM] Could you specify the types of devices and network conditions we should consider for ensuring acceptable opening and download times?

[USER] We need to ensure that our application performs well across a variety of devices and network conditions. Specifically, we should consider opening times and download times for both desktops (Windows 10, macOS Catalina) and mobile devices (iOS 14+, Android 9+). The system should be able to handle at least 1000 concurrent users without significant degradation in performance. Ideally, pages should open within less than 2 seconds under normal network conditions. For slower networks or lower-end devices, we expect the opening time to be no more than 5 seconds. It's important that our application can handle a range of internet speeds from fast Wi-Fi to slow 3G connections.

[SYSTEM] Could you specify the acceptable downtime periods and incident limits for our system during critical operations or maintenance windows to ensure we meet your reliability requirements?

[USER] For our system, we need the planned downtime to be kept to a minimum. Ideally, it should not exceed 8 hours in any rolling three-month period. Unplanned downtime is also critical; it must not happen more than twice during that same period. When an incident does occur, whether due to software or hardware failure, we expect the system to be restored within no more than 4 hours, with inline synchronization as part of the recovery process.

[SYSTEM] Could you provide more details on how the role-based permissions will be enforced for different types of searches and what specific actions users in each role should be able to perform?

[USER] For role-based permissions, we need to ensure that when a user performs a quick or advanced search, the system never includes any records in the result list that they don't have access to. This is important for maintaining data security and privacy. Additionally, it should be possible for users to belong to more than one group, allowing for flexible role assignments based on their tasks and responsibilities. When designing different access paths or navigation structures for various user groups, we need to consider the specific tasks and information needs of each group to ensure they can efficiently find what they need.

Part C - Extracted requirements by category

Functional

- FR-001** For placing an order, we should streamline the process so that customers can add items to their cart, proceed to checkout with minimal steps, and complete payment without too many interactions.
- FR-002** Similarly, checking order status should only require logging in or entering an order number, then viewing the current status page which displays all relevant information like delivery date and tracking details.
- FR-003** For example, when a customer logs in and starts a new order, it should recognize them and auto-populate fields like shipping address and payment method where applicable.
- FR-004** This will save them time and reduce errors.
- FR-005** Ideally, it would also allow us to show any saved payment details securely so they can just confirm or change if needed.
- FR-006** It should be able to manage a list of items across different orders, so that if they start one order but then want to make another, they can easily switch between them without losing any items.
- FR-007** For role-based permissions, we need to ensure that when a user performs a quick or advanced search, the system never includes any records in the result list that they don't have access to.
- FR-008** Additionally, it should be possible for users to belong to more than one group, allowing for flexible role assignments based on their tasks and responsibilities.
- FR-009** When designing different access paths or navigation structures for various user groups, we need to consider the specific tasks and information needs of each group to ensure they can efficiently find what they need.

Non-Functional

- NFR-001** It's really important that users can customize their menu contents based on their role or preferences.
- NFR-002** We need to ensure the page length is appropriate for each task, and we should use color carefully, considering accessibility issues.
- NFR-003** Familiar terms are key — navigation links should be intuitive, using language that users will recognize from their experience with similar systems.
- NFR-004** For example, graphical images or videos should have alternative descriptions provided.
- NFR-005** This is important for accessibility and ensuring everyone can understand the content.
- NFR-006** We also want to make sure our user interfaces allow activation of controls through a variety of input devices, not just keyboards or mice.
- NFR-007** Currently, some of our pages have a splash screen which can be an issue because it might not always allow for easy navigation using these tools.

NFR-008	Ideally, we should offer a skip link to bypass the splash screen so they can get straight to the content they need.
NFR-009	We also want to avoid horizontal scrolling as much as possible because it can be difficult for users with visual impairments to navigate using screen readers.
NFR-010	Lastly, we should ensure that navigation components like menus and links are placed consistently across the application so that users can predict where to find important features.
NFR-011	For example, when reading long articles or filling out forms, being able to increase the text size can make it much easier on the eyes and improve readability.
NFR-012	We should ensure that when users navigate through different sections of the application, they don't encounter issues like broken back button functionality or bookmarking problems.
NFR-013	Currently, the system can be a bit cumbersome because it has too many links that make the text hard to read at a glance.
NFR-014	We should provide an alternative text-only page as well, which would allow them to skim without getting distracted by hyperlinks.
NFR-015	To support skimming better, we need clear links, bulleted lists, highlighted keywords, and logical headings.
NFR-016	The links themselves should be marked with dashed underlines instead of solid ones to avoid overwhelming the user.
NFR-017	It's important that our interface follows a consistent design so users can easily navigate and find what they're looking for without feeling lost or confused.
NFR-018	We need to ensure that users can easily find help when they need it.
NFR-019	Each page should have a link back to the home or a recognizable landmark page.
NFR-020	To make sure users can understand the text quickly, we'll use short sentences, break up long texts into chunks, and present information as bullet points.
NFR-021	We need to ensure that users can easily understand where they are in the application.
NFR-022	For example, when a user is filling out a form for creating an account, it should clearly indicate which step they're on and how many steps remain.
NFR-023	spelling and grammar must be impeccable so as not to impede readability.
NFR-024	For instance, screen readers should be able to navigate through the application smoothly without any issues.
NFR-025	Ideally, it would be great if we could offer a variety of support features both within the app and through a browser interface.
NFR-026	If someone needs help with something specific, they should be able to access that information quickly without having to navigate away from their current task.
NFR-027	Each page should provide this information clearly and succinctly.
NFR-028	Currently, we've had issues with too many clicks required to get to certain sections, which can be frustrating for everyone but especially problematic for those with cognitive disabilities.
NFR-029	We should aim to streamline navigation as much as possible.
NFR-030	This will help create a familiar and common look and feel, making it easier for all users to navigate.
NFR-031	For example, when a link opens a new window or tab, it includes an icon next to the text to indicate this, ensuring that all users can easily identify such links.
NFR-032	For example, a user with limited permissions might try to view or modify a case they do not have access to.
NFR-033	Logging these attempts is crucial because it helps us track who tried what and when, which is vital for security audits.

NFR-034	By exporting audit trails for specific cases without affecting the stored data, we can review these logs later if needed.
NFR-035	The audit trail must be unalterable—no one should be able to change or delete it, ensuring its integrity.
NFR-036	For user profiles in our application, we need a way for users to log any defects or enhancement requests and track them.
NFR-037	Regarding profile management, only super-users should be able to change security attributes like access rights, security levels, privileges, and password management.
NFR-038	Users should have the ability to see, modify, and delete their own profiles as needed.
NFR-039	This ensures that each user can manage their own information while maintaining a high level of security through strict control over who can make changes.
NFR-040	For cases accessed within the last two months, it should be able to retrieve and display them within 5-8 seconds — this is crucial for a good user experience.
NFR-041	For older cases not accessed in the past two months, retrieval and display should happen within 20 seconds.
NFR-042	This ensures that while less frequently used data can take longer to load, it still meets our needs without causing significant delays.
NFR-043	Specifically, we should consider opening times and download times for both desktops (Windows 10, macOS Catalina) and mobile devices (iOS 14+, Android 9+).
NFR-044	The system should be able to handle at least 1000 concurrent users without significant degradation in performance.
NFR-045	Ideally, pages should open within less than 2 seconds under normal network conditions.
NFR-046	It's important that our application can handle a range of internet speeds from fast Wi-Fi to slow 3G connections.
NFR-047	Ideally, it should not exceed 8 hours in any rolling three-month period.
NFR-048	it must not happen more than twice during that same period.

Data

DATA-001	For example, we allow users to define their own default values for certain fields, which is really helpful when they need to enter similar information repeatedly.
DATA-002	Sometimes, error messages can be unclear or too technical, leading to confusion among our users.
DATA-003	We've had issues where the system doesn't provide clear instructions on how to correct the input, making it difficult for them to resolve errors themselves.
DATA-004	Additionally, while persistent defaults are useful, there have been times when they caused problems because users forgot to update their settings.
DATA-005	We need a way to ensure that these defaults can be easily changed or overridden if necessary.
DATA-006	However, we've had issues with data validation when these defaults don't align with our international standards.
DATA-007	For example, AUTO-0082 requires us to consult specific International Standards for guidance, but sometimes user-defined defaults can conflict with those guidelines.
DATA-008	AUTO-0001 emphasizes the importance of following these procedures, so we must find a balance between user convenience and strict adherence to regulations.
DATA-009	Additionally, we've noticed that relying solely on color coding can be problematic, as it should never be the only means of conveying important information.
DATA-010	We need clear and specific error messages when user-defined defaults conflict with international standards.

DATA-011 For example, if a user sets a date format that doesn't match the ISO standard, we should inform them about the correct format and suggest using something like YYYY-MM-DD.

Interfaces

- IF-001** For example, our users should be able to adjust settings according to their needs, making sure everything remains intuitive and easy to use for both end-users and administrators.
- IF-002** The design must follow broad guidelines such as supporting multilingual interfaces, which will cater to a diverse user base.
- IF-003** Additionally, the system should ensure that no operational data is lost in case of equipment or communication network failures.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
Registration Citizens can register their complaints with police and then based on the evidence, facts and following investigation, police shall take the complaint forward.	X	— not recovered
Help Module The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	X	— not recovered
The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	X	— not recovered
The solution should provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.	✓	For user profiles in our application, we need a way for users to log any defects or enhancement requests and track them.
The solution should send alerts (e.g., email, SMS) to the user if the user chooses to whenever any action has been taken on the alert.	X	— not recovered
The solution should enable the user to track the submitted defect or enhancement request.	X	— not recovered
The solution should enable the help-desk user to view the reports on the submitted defects or enhancement requests category-wise, status-wise, and age- wise.	X	— not recovered
The support solution should be accessible to the users both from within the application and also outside the application through a browser interface.	✓	Ideally, it would be great if we could offer a variety of support features both within the app and through a browser interface.
The System must keep an unalterable audit trail capable of automatically capturing and storing information about: All the actions (create/read/update/delete) that are taken upon the critical entities (case, suspect, property,...) in the system The user initiating and or carrying out the action;	X	— not recovered
Administrative parameters The word “unalterable” is to mean that the audit trail data cannot be modified in any way or deleted by any user;	✓	The audit trail must be unalterable—no one should be able to change or delete it, ensuring its integrity.
it may be subject to re-department and copying to removable media if required, so long as its contents remain unchanged.	X	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Once the audit trail functionality has been activated, the System must track events without manual intervention, and store in the audit trail information about them.	✗	— not recovered
The System must maintain the audit trail for as long as required, which will be at least for the life of the case to which it refers.	✗	— not recovered
The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	✓	By exporting audit trails for specific cases without affecting the stored data, we can review these logs later if needed.
The System must be able to capture and store violations (i.e.	✓	Additionally, the system should ensure that no operational data is lost in case of equipment or communication network failures.
The System must at a minimum be able to provide reports for actions on cases organised: By case;	✗	— not recovered
The System should be able to provide reports for actions on cases organised by workstation and (where technically appropriate) by network address.	✗	— not recovered
The System must allow the user to limit access to cases to specified users or user groups.	✓	For example, a user with limited permissions might try to view or modify a case they do not have access to.
The system should provide for role-based control for the functionality within the system The System must allow a user to be a member of more than one group.	✓	Additionally, it should be possible for users to belong to more than one group, allowing for flexible role assignments based on their tasks and responsibilities.
The System must allow only admin-users to set up user profiles and allocate users to groups.	✗	— not recovered
The System should allow a user to stipulate which other users or groups can access cases.	✗	— not recovered
The System must allow changes to security attributes for groups or users (such as access rights, security level, privileges, password allocation and management) to be made only by super-user.	✓	Regarding profile management, only super-users should be able to change security attributes like access rights, security levels, privileges, and password management.
If a user requests access to, or searches for, a case which he does not have the right to access, the System must provide one of the following responses (selectable at configuration time): display title and metadata;	✗	— not recovered
the most stringent) implies that the System must not include such cases in any count of search results;	✗	— not recovered
If a user performs a quick or advanced search, the System must never include in the search result list any record which the user does not have the right to access.	✓	For role-based permissions, we need to ensure that when a user performs a quick or advanced search, the system never includes any records in the result list that they don't have access to.
If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.	✓	Logging these attempts is crucial because it helps us track who tried what and when, which is vital for security audits.
Any access to cases, and all other activities involving the cases and related documents or data should also need to be stored in the audit trail to ensure legal admissibility and to assist in data recovery.	✗	— not recovered
Ease of Use All error messages produced by the System must be meaningful, so that they can be appropriately acted upon by the users who are likely to see them.	✓	Sometimes, error messages can be unclear or too technical, leading to confusion among our users.

Source requirement text	Matched?	Closest generated requirement text
The System must employ a single set of user interface rules, or a small number of sets to provide a familiar and common look and feel for the application.	✗	— not recovered
The System must be able to display several entities (cases, suspects) simultaneously.	✗	— not recovered
The interfaces must be made customizable or user-configurable to the extent possible.	✗	— not recovered
Such configurations must be saved in the user profile.	✗	— not recovered
The System user interface must be suitable for users with special needs;	✗	— not recovered
that is, compatible with specialist software that may be used and with appropriate interface guidelines The System must provide End User and Administrator functions which are easy to use and intuitive throughout.	✓	For example, our users should be able to adjust settings according to their needs, making sure everything remains intuitive and easy to use for both end-users and administrators.
The System must allow persistent defaults for data entry where desirable.	✓	Additionally, while persistent defaults are useful, there have been times when they caused problems because users forgot to update their settings.
These defaults should include: user-definable values;	✗	— not recovered
Frequently-executed System transactions must be designed so that they can be completed with a small number of interactions (e.g.	✗	— not recovered
Where the System employs a graphical user interface, it must allow users to customise it.	✗	— not recovered
Customisation should include, but need not be limited to the following changes: menu contents;	✓	It's really important that users can customize their menu contents based on their role or preferences.
Usability The user interfaces should be designed to make them user-intuitive.	✓	It's important that our interface follows a consistent design so users can easily navigate and find what they're looking for without feeling lost or confused.
The user interfaces of the system should comply with Standard ISO 9241.	✗	— not recovered
ICT accessibility: ISO 9241-20 shall be the standard for guidance on ICT accessibility.	✗	— not recovered
Software accessibility ISO 9241-171 shall be the standard for guidance on software accessibility.	✗	— not recovered
User interfaces should meet its requirements and recommendations.	✗	— not recovered
Content accessibility WCAG 1.0 shall be the standard used for guidance on content accessibility.	✓	This is important for accessibility and ensuring everyone can understand the content.
Providing text equivalents for non-text media objects: All non-text media objects, such as graphical images or video, should be provided with alternative equivalent textual descriptions and/or with equivalent text-based functionality.	✓	For example, graphical images or videos should have alternative descriptions provided.
Making navigation self-descriptive: Navigation should be designed to help users understand where they are, where they have been and where they can go next.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Showing users where they are: Each presentation segment (page or window) should provide the user with a clear and sufficient indication of where he or she is in the navigation structure and of the current segment position with respect to the overall structure.	✗	— not recovered
Offering alternative access paths: Alternative access paths for navigating to a specific unit of content should be offered to support different navigation strategies.	✗	— not recovered
Minimizing navigation effort: The number of navigation steps needed to reach a certain piece of content should be minimized as long as different mental models, navigation strategies and tasks of the user are taken into account.	✗	— not recovered
Splash screens should be avoided unless they provide useful content or feedback about the application state to the user.	✓	Currently, some of our pages have a splash screen which can be an issue because it might not always allow for easy navigation using these tools.
If a splash screen is used, a navigation option to skip it should be offered.	✓	Ideally, we should offer a skip link to bypass the splash screen so they can get straight to the content they need.
Avoiding opening unnecessary windows: Additional windows such as new browser windows or pop-up windows should only be opened if this supports the user's task.	✗	— not recovered
Horizontal scrolling should be avoided wherever possible.	✓	We also want to avoid horizontal scrolling as much as possible because it can be difficult for users with visual impairments to navigate using screen readers.
Designing for input device independence: User interfaces should be designed to allow activation of controls by a variety of input devices.	✓	We also want to make sure our user interfaces allow activation of controls through a variety of input devices, not just keyboards or mice.
Making user interfaces robust: User interfaces should be designed to be as robust as possible in the face of changing technology.	✗	— not recovered
Acceptable opening / download times: Application pages should be designed and implemented so that there are acceptable opening times and download times for the expected range of technical contexts of use (e.g.	✓	Ideally, pages should open within less than 2 seconds under normal network conditions.
Minimizing user errors: Potential user errors as well as the effort needed to recover from errors should be minimized.	✓	This will save them time and reduce errors.
Providing clear error messages: The content of error messages shown on the pages or special error pages should clearly state the reason why the error occurred and, if possible, actions the user can take to resolve the error.	✗	— not recovered
Making text resizable by the user: Text should be able to be resized by the user, using functions provided by the user agent or other appropriate means i.e.	✗	— not recovered
Text quality: The quality of textual content with respect to spelling and grammar should be sufficient so as not to impede readability.	✓	spelling and grammar must be impeccable so as not to impede readability.

Source requirement text	Matched?	Closest generated requirement text
Writing style: The reading and understanding of the textual content on the screen should be supported by suitable means, including the use of short sentences, the division of the text into shorter chunks or the presentation of content items in the form of bullet points.	✓	To make sure users can understand the text quickly, we'll use short sentences, break up long texts into chunks, and present information as bullet points.
Supporting text skimming: Fast skimming of text should be supported by the provision of clear links, bulleted lists, highlighted keywords, logical headings, and short phrases and sentences.	✓	To support skimming better, we need clear links, bulleted lists, highlighted keywords, and logical headings.
Readability of text: Text presented on the pages should be readable taking into account the expected display characteristics and spatial arrangement.	✗	— not recovered
ISO 9241-303 shall be consulted for screen text legibility requirements.	✗	— not recovered
Distinguishable within-page links: Within-page links should be clearly distinguishable from other links that lead to a different page.	✗	— not recovered
Within-page links are shown with dashed rather than solid underlines Avoiding link overload: Text pages containing large proportions of links should be formatted so that the presence of links does not impede the readability of the text.	✓	The links themselves should be marked with dashed underlines instead of solid ones to avoid overwhelming the user.
Using familiar terminology for navigation links: Navigation links — particularly links representing the main navigation structure — should be labelled with terms that are familiar to the user, based on his/her general knowledge, prior experience in the application domain or experience of using other systems.	✓	Familiar terms are key — navigation links should be intuitive, using language that users will recognize from their experience with similar systems.
Marking links opening new windows: Links that open new browser windows or pop-up windows should be clearly marked.	✓	For example, when a link opens a new window or tab, it includes an icon next to the text to indicate this, ensuring that all users can easily identify such links.
Distinguishing navigation links from controls: Navigation links should be clearly distinguishable from controls activating some action.	✗	— not recovered
space filled only with the background color should be used in such a way that it does not impair the visual skimming of the page.	✓	We need to ensure the page length is appropriate for each task, and we should use color carefully, considering accessibility issues.
Selecting appropriate page lengths The length of a page should be selected so as to support the primary purpose and use of the page.	✗	— not recovered
Using colour: Colour should be used with care, taking into account human capabilities and restrictions in perceiving colour, and not as the only means of conveying information.	✗	— not recovered
Color should never be the only means of coding.	✓	Additionally, we've noticed that relying solely on color coding can be problematic, as it should never be the only means of conveying important information.
Using frames with care: If frames are used, care should be taken to avoid possible problems, for example, those involving the use of the back button, bookmarking of pages, or scrolling of information.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Providing alternatives to frame-based presentation: If frames are used, an alternative way of presenting relevant information without frames should be provided.	✗	— not recovered
Providing alternative text-only pages: When style sheets and/or frames are turned off it should be possible for the user to read and understand the page;	✗	— not recovered
alternatively, the user should be provided with an equivalent alternative text-only page.	✓	We should provide an alternative text-only page as well, which would allow them to skim without getting distracted by hyperlinks.
Consistent page layout: Pages should be designed using consistent layout schemes, supporting the user in finding similar information at the same position on different pages.	✗	— not recovered
Placing title information consistently: Page titles should be placed in a consistent location on the different pages.	✗	— not recovered
Observing principles of human perception When designing application pages, the general principles of human perception should be taken into account.	✗	— not recovered
The International Standards mentioned below shall be consulted for guidance.	✓	For example, AUTO-0082 requires us to consult specific International Standards for guidance, but sometimes user-defined defaults can conflict with those guidelines.
In addition, when designing multimedia information presentations, the design principles and recommendations described in ISO 14915-1 to ISO 14915-3 should be taken into account.	✗	— not recovered
Linking back to the home page or landmark pages: Each page should contain a link leading to the home page of the application or to a landmark page that is easy to recognize for the user.	✓	Each page should have a link back to the home or a recognizable landmark page.
Providing a site map: A separate navigation overview such as a site map should be provided for application showing the structure of the site in an overview form.	✗	— not recovered
Consistency between navigation components and content: If navigation components (or overviews) are shown in conjunction with associated content, consistency between the navigation component and the content shown should be maintained by indicating in the navigation component (e.g.	✗	— not recovered
Placing navigation components consistently: Navigation components should be placed consistently on the pages or in the framesets in the pages of the application.	✓	Lastly, we should ensure that navigation components like menus and links are placed consistently across the application so that users can predict where to find important features.
Taking account of the users' tasks and information needs: When providing different access paths or navigation structures for different user groups, the tasks and information needs of these user groups should be taken into consideration.	✓	When designing different access paths or navigation structures for various user groups, we need to consider the specific tasks and information needs of each group to ensure they can efficiently find what they need.
Making individualization and adaptation evident: It should be made evident to the user when individualization and/or adaptation are used.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Making user profiles evident: If predefined user profiles or user-specified profiles are used for individualizing or adapting content, the profile currently used should be made evident.	✗	— not recovered
Allowing users to see and change profiles: If user-specified profiles are used, users should be able to see, modify and delete that profile on demand.	✓	Users should have the ability to see, modify, and delete their own profiles as needed.
The user interfaces of the system should follow the guidelines specified under www.usability.gov System Availability The System must be available to users: from <xx:00> to <xx:00>;	✗	— not recovered
The planned downtime for the System must not exceed <xx> hours per <rolling three month period>.	✗	— not recovered
Unplanned downtime for the System must not exceed <xx hours/minutes> per <rolling three month period>.	✓	Ideally, it should not exceed 8 hours in any rolling three-month period.
The number of incidents of unplanned downtime for the System must not exceed <x> per <rolling three month period>.	✗	— not recovered
In the event of any software or hardware failure, it must be possible to restore the System (with inline synchronization) within no more than <xx> hours.	✗	— not recovered
The System must be able to retrieve and display within 5-8 seconds the case which has been accessed within the previous 2 months, regardless of storage capacity or number of cases in the system.	✓	For cases accessed within the last two months, it should be able to retrieve and display them within 5-8 seconds — this is crucial for a good user experience.
The System must be able to retrieve and display within 20 seconds the case which has not been accessed within the previous 2 months, regardless of storage capacity or number of cases in the system.	✓	For older cases not accessed in the past two months, retrieval and display should happen within 20 seconds.
The System be scaleable and must not have any features which would preclude use in small or large police stations, with varying numbers of cases handled.	✗	— not recovered
The systems should be designed with the following broad guidelines: System Functionality The system should support multilingual interface The system should be designed in manner that operational data is not lost in case of any failure of equipment or communication network.	✓	The design must follow broad guidelines such as supporting multilingual interfaces, which will cater to a diverse user base.

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.2535 / 0.1800 / 0.2105. Weighted P/R/F1: 0.4930 / 0.3500 / 0.4094.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	Taking account of the users' tasks and information needs: When providing different access paths or navigation structures for different user groups, the tasks and information needs of these user groups should be taken into consideration.	When designing different access paths or navigation structures for various user groups, we need to consider the specific tasks and information needs of each group to ensure they can efficiently find what they need.	The generated requirement accurately reflects the source's need to consider user tasks and information needs when designing navigation for different user groups, preserving all critical details.

Verdict	Source requirement	Generated requirement	Judge reason
full	Allowing users to see and change profiles: If user-specified profiles are used, users should be able to see, modify and delete that profile on demand.	Users should have the ability to see, modify, and delete their own profiles as needed.	The generated requirement faithfully covers the ability for users to see, modify, and delete their own profiles, aligning with the source's intent for user-specified profiles on demand.
full	If a user performs a quick or advanced search, the System must never include in the search result list any record which the user does not have the right to access.	For role-based permissions, we need to ensure that when a user performs a quick or advanced search, the system never includes any records in the result list that they don't have access to.	The generated requirement faithfully covers the same core need and preserves all critical details regarding excluding unauthorized records from search results. The phrase 'For role-based permissions' adds context but does not alter the core requirement.
full	Linking back to the home page or landmark pages: Each page should contain a link leading to the home page of the application or to a landmark page that is easy to recognize for the user.	Each page should have a link back to the home or a recognizable landmark page.	The generated requirement faithfully captures all critical details of the source, including the presence of a link on each page to either the home page or a recognizable landmark page.
full	Using familiar terminology for navigation links: Navigation links — particularly links representing the main navigation structure — should be labelled with terms that are familiar to the user, based on his/her general knowledge, prior experience in the application domain or experience of using other systems.	Familiar terms are key — navigation links should be intuitive, using language that users will recognize from their experience with similar systems.	The generated requirement faithfully captures the essence of using familiar and intuitive terminology for navigation links, referencing user experience with similar systems, which aligns with the source's details.
full	The solution should provide an interface for the user to log any defects or enhancement requests on the application and track them thereafter.	For user profiles in our application, we need a way for users to log any defects or enhancement requests and track them.	The generated requirement faithfully covers the core need of providing a way for users to log and track defects/enhancement requests, preserving all critical details. The mention of "user profiles" provides context but does not materially alter the requirement.
full	Horizontal scrolling should be avoided wherever possible.	We also want to avoid horizontal scrolling as much as possible because it can be difficult for users with visual impairments to navigate using screen readers.	The generated requirement faithfully covers the source, preserving the core need and modality, and the added reason does not constitute a material addition.
full	The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	By exporting audit trails for specific cases without affecting the stored data, we can review these logs later if needed.	The generated requirement faithfully covers the ability to export audit trails for specific cases without altering stored data. The added phrase about reviewing logs later is a consequence, not a material addition or alteration of the core requirement.
full	Within-page links are shown with dashed rather than solid underlines Avoiding link overload: Text pages containing large proportions of links should be formatted so that the presence of links does not impede the readability of the text.	The links themselves should be marked with dashed underlines instead of solid ones to avoid overwhelming the user.	The generated requirement accurately captures the specific detail about using dashed underlines for links instead of solid ones, which is the first part of the source requirement.

Verdict	Source requirement	Generated requirement	Judge reason
full	Designing for input device independence: User interfaces should be designed to allow activation of controls by a variety of input devices.	We also want to make sure our user interfaces allow activation of controls through a variety of input devices, not just keyboards or mice.	The generated requirement faithfully captures the need for user interfaces to support various input devices, preserving all critical details.
full	The system should provide for role-based control for the functionality within the system The System must allow a user to be a member of more than one group.	Additionally, it should be possible for users to belong to more than one group, allowing for flexible role assignments based on their tasks and responsibilities.	The generated requirement faithfully covers the second part of the source requirement, stating that users can belong to more than one group. The additional explanation about flexible role assignments does not alter the core requirement.
full	Placing navigation components consistently: Navigation components should be placed consistently on the pages or in the framesets in the pages of the application.	Lastly, we should ensure that navigation components like menus and links are placed consistently across the application so that users can predict where to find important features.	The generated requirement faithfully covers the consistent placement of navigation components across the application, preserving all critical details. The added explanation of the benefit does not alter the core requirement.
full	Administrative parameters The word “unalterable” is to mean that the audit trail data cannot be modified in any way or deleted by any user;	The audit trail must be unalterable—no one should be able to change or delete it, ensuring its integrity.	The generated requirement accurately defines 'unalterable' for the audit trail, matching the source's definition of not being modifiable or deletable by any user.
full	Writing style: The reading and understanding of the textual content on the screen should be supported by suitable means, including the use of short sentences, the division of the text into shorter chunks or the presentation of content items in the form of bullet points.	To make sure users can understand the text quickly, we'll use short sentences, break up long texts into chunks, and present information as bullet points.	The generated requirement accurately captures the core need and all specific examples of writing style from the source.
full	The System must allow persistent defaults for data entry where desirable.	For example, we allow users to define their own default values for certain fields, which is really helpful when they need to enter similar information repeatedly.	The generated requirement faithfully covers the allowance of persistent defaults for data entry, using an example that clarifies the 'desirable' aspect.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
Help Module The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	We need to ensure that users can easily find help when they need it.	The generated requirement is too general, losing critical details about the help being 'detailed,' 'context-sensitive,' and covering 'all possible actions and scenarios on all user interfaces.'
The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	If someone needs help with something specific, they should be able to access that information quickly without having to navigate away from their current task.	The generated requirement captures the core need for quick, context-sensitive help access but loses the critical detail about the scope of the help material (all actions/scenarios on all UIs) and the explicit mention of 'detailed material'.

Source requirement	Generated requirement	Judge reason
The solution should enable the user to track the submitted defect or enhancement request.	For user profiles in our application, we need a way for users to log any defects or enhancement requests and track them.	The generated requirement covers tracking defects/enhancement requests but also adds the requirement to provide a way for users to log them, which was not explicitly stated in the source.
The System must keep an unalterable audit trail capable of automatically capturing and storing information about: All the actions (create/read/update/delete) that are taken upon the critical entities (case, suspect, property,...) in the system The user initiating and or carrying out the action;	The audit trail must be unalterable—no one should be able to change or delete it, ensuring its integrity.	The generated requirement only covers the 'unalterable' aspect of the audit trail, omitting the critical details about the specific information (actions, entities, user) that must be captured and stored.
If a user requests access to, or searches for, a case which he does not have the right to access, the System must provide one of the following responses (selectable at configuration time): display title and metadata;	For role-based permissions, we need to ensure that when a user performs a quick or advanced search, the system never includes any records in the result list that they don't have access to.	The source specifies providing a specific response (display title and metadata) for unauthorized access/search, while the generated text states that unauthorized records should never be included in the search result list, which is a different system behavior.
the most stringent) implies that the System must not include such cases in any count of search results;	For role-based permissions, we need to ensure that when a user performs a quick or advanced search, the system never includes any records in the result list that they don't have access to.	The source specifically states that unauthorized cases must not be included in any 'count of search results', a critical detail that is not explicitly preserved by the generated text's focus on the 'result list'.
Any access to cases, and all other activities involving the cases and related documents or data should also need to be stored in the audit trail to ensure legal admissibility and to assist in data recovery.	The audit trail must be unalterable—no one should be able to change or delete it, ensuring its integrity.	The generated requirement focuses on the unalterability of the audit trail, which is a material addition and a different aspect than the source's focus on what activities must be stored and for what purpose (legal admissibility, data recovery).
Ease of Use All error messages produced by the System must be meaningful, so that they can be appropriately acted upon by the users who are likely to see them.	This will save them time and reduce errors.	The generated text states a general benefit ("save them time and reduce errors") without specifying the requirement for meaningful and actionable error messages.
Ease of Use All error messages produced by the System must be meaningful, so that they can be appropriately acted upon by the users who are likely to see them.	We've had issues where the system doesn't provide clear instructions on how to correct the input, making it difficult for them to resolve errors themselves.	The generated text describes a specific issue with unclear error instructions, but does not state the requirement that error messages *must* be meaningful and actionable.
The System must be able to display several entities (cases, suspects) simultaneously.	For cases accessed within the last two months, it should be able to retrieve and display them within 5-8 seconds — this is crucial for a good user experience.	The generated requirement focuses on retrieval speed for a subset of cases, losing the critical detail of displaying multiple entities simultaneously and adding new constraints.

Document: 0000_gamma_j

Part A - Document header

Document ID	0000_gamma_j
Generation model	ollama / qwen2.5:7b-instruct
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	57
Generated requirement count	49
Matched count	21
Clarification rounds	requested 2 / used 0
Clarification chunks	0
Semantic P / R / F1	0.4286 / 0.3684 / 0.3962
LLM strict P / R / F1	0.1837 / 0.1579 / 0.1698
LLM weighted P / R / F1	0.3878 / 0.3333 / 0.3585
Evidence / Propositions / Gap	88 / 49 / 0
Grounded after rewrite	49

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 0000 - gamma j system, and the goal is to capture the main requirements and constraints for document 0000_gamma_j.

[SYSTEM] Could you provide more details on how the system should handle different types of items in the shopping cart?

[USER] We need the system to ensure that customers always have items in their shopping cart. It should be able to handle different types of products and quantities. For security, we want it to run test credit card transactions before processing real payments. Yoggie should work closely with our organization on future enhancements and new features. The system must exhibit specific functionalities under certain conditions, which is important for us because it ensures that all the necessary features are covered.

[SYSTEM] Can you provide more details on the types of notifications that need to be sent via email to customers and vendors, and under what conditions these emails should be triggered?

[USER] We need the system to send emails to customers and vendors as quickly as possible — ideally in less than a second. This is important because we want to ensure that notifications are received promptly, which helps maintain trust and satisfaction with our service. For both customers and vendors, these emails should be triggered under specific conditions such as order confirmations, payment reminders, account updates, and support requests. We also need the system to handle large volumes of concurrent users — up to 1000 at a time — without delay or failure.

[SYSTEM] Could you provide more details on the specific functionalities that need to be enabled under certain conditions to ensure all necessary features are covered?

[USER] We need the system to handle a minimum of 1000 concurrent users without any performance degradation. For our e-commerce platform, it should be able to process transactions quickly — ideally within less than 2 seconds from when a customer clicks 'buy now'. Security is a big concern for us; customer data needs to be encrypted both in transit and at rest. We've had issues with slow login times, so we need the system to authenticate users efficiently, ensuring that it never takes more than 5 seconds for a user to log in. For returning customers, the system should remember their card details but verify the card is still valid upon each login. It's important to us because we want to reduce friction and increase customer satisfaction. Additionally, the system must support multiple browsers including Chrome, Firefox, Safari, and Edge on both Windows and macOS operating systems. We need it to be compatible with our existing server infrastructure, specifically using Apache for web servers and Nginx for reverse proxying. The user interface should display product titles and metadata prominently but never show any private information like credit card numbers or passwords in the browser interface. Only authorized users, identified by their unique login credentials, should have access to certain administrative features. We must not allow unauthorized users to modify or view sensitive data. Lastly, we need robust logging and auditing capabilities for all user activities, ensuring that every action is recorded and can be traced back to a specific user.

[SYSTEM] Can you provide more details on how the system should handle potential security breaches or vulnerabilities identified during a security audit?

[USER] Security is a top priority for us. To protect the system, we need it to be safeguarded by an open-source firewall called Firestarter. Additionally, the system should automatically detect and block IP addresses that show signs of DOS attacks. We also want all sensitive information to be encrypted using https to prevent any breaches. Lastly, we need the system to monitor login attempts; if someone tries to log in multiple times without success, it should lock them out after a certain number of failed attempts.

[SYSTEM] Could you provide more details on the criteria for validating email addresses and how often this validation should occur to ensure accuracy?

[USER] We need the system to validate email addresses for accuracy. Ideally it would check existence and format of emails before saving them in our SQL-based database, which is crucial for storing inventory information. This validation should occur every time a new email address is entered or an existing one is updated. While we're focused on email validation, we also have another requirement to automatically compress image files that are too large — this needs to happen without affecting the quality of the images. The system must use a SQL-based database for both inventory and other data storage to ensure consistency and reliability.

[SYSTEM] Could you provide more details on the specific conditions under which emails to customers and administrators should be sent, and what information should each email contain?

[USER] For customer emails, we need the system to send an email confirmation that includes the items ordered and their tracking number. This should happen automatically after the order is placed. For administrators, it's important that they receive notifications about any technical queries from customers or sales people. The web interface must be consistent across different browsers like Chrome, Firefox, and Safari, ensuring a user-friendly experience for our customers. Security is key, so all communications need to use HTTPS.

[SYSTEM] Could you specify the exact conditions under which the system should send emails to customers and administrators, including the content that each email should contain?

[USER] For customer support queries, the system should send an email to our System Administrator whenever a technical issue is raised by customers or sales staff. The content of these emails should include details like the user's contact information and the nature of the problem so that it can be addressed promptly. Regarding order confirmations, the WebOrder system needs to notify customers via email once their items are being shipped. This email should contain the tracking number for easy reference. We operate on an Intel-based system with Slackware Linux 2.6 and Apache Web Server, ensuring a stable environment. Customers must access the web order system through a secure HTTPS connection from any compatible browser.

[SYSTEM] Could you clarify the specific details on how the system should handle email confirmations for customers, such as the timing and any additional information that might be included?

[USER] We need the system to send an email confirmation to customers as soon as their order is placed. This email should include details about the items they ordered and the shipping address, along with a tracking number for easy reference. It's important because it helps ensure that customers are aware of their order status and can track their deliveries.

[SYSTEM] Could you provide more details on the specific scenarios where the system should perform these efficiency requirements, such as during peak usage times or when handling a large number of concurrent users?

[USER] During peak usage times, such as when we have a large number of concurrent users, the system should be able to handle searches efficiently. For example, it should use Dijkstra's shortest path algorithm for searching, and it needs to perform these searches quickly — ideally in less than 1 second per search. Additionally, when dealing with a high volume of data, like restoring records or retrieving products, we need the system to maintain its speed. Specifically, it should be able to restore 1000 records per second and retrieve up to 200 products per second without any significant slowdowns.

[SYSTEM] Can you specify the exact conditions under which the system should deploy and become operational after a USB is plugged in, such as during initial setup or after maintenance?

[USER] During initial setup or after maintenance, we need the system to be able to deploy and become operational within less than a minute upon plugging in the USB. This is crucial for minimizing downtime and ensuring smooth transitions.

[SYSTEM] Could you provide more details on the specific conditions under which the system should run test shipping orders, such as the types of tests and their frequency?

[USER] We need the system to be able to run test shipping orders regularly. Ideally it would happen automatically at least once a day, but we should also have the option to manually trigger these tests whenever necessary. The types of tests include verifying that all order information is correctly processed and that the shipping details are accurate. It's important because we want to ensure everything works smoothly before actual customer orders go through. We've had issues with incorrect shipping in the past, so frequent testing helps catch those problems early.

[SYSTEM] Could you specify the exact conditions under which the system should send out automated test emails to ensure email functionality is working correctly, such as during scheduled maintenance or after software updates?

[USER] For sure, we need the system to send out automated test emails during scheduled maintenance and after software updates. This is important because it ensures that our email functionality remains reliable. During maintenance windows, say between 8 PM and 2 AM on Tuesday nights, the system should automatically check if all outgoing emails are working properly. Similarly, right after a new version of the software is deployed, we want to make sure everything is still functioning as expected, so the system should send out test emails then too. This way, any issues can be caught early and addressed before they affect our users.

[SYSTEM] Could you specify the exact conditions under which the system should initiate and complete a backup of critical data, such as during off-peak hours or before performing maintenance?

[USER] Ideally it would initiate a backup during off-peak hours to minimize any impact on our users. Specifically, the system should start backing up critical data between 2 AM and 6 AM local time. This is important because we have had issues with backups interfering with user experience during regular business hours. Additionally, before performing maintenance tasks such as software updates or server checks, it should complete a backup to ensure that any changes are safe and can be reverted if necessary. We need this to happen automatically so our IT team doesn't have to manually trigger the process every time.

[SYSTEM] Could you clarify the specific permissions and actions a Sales Person should be able to perform within the Weborder System after their account is registered and they are logged in?

[USER] Sales people need to be able to place orders and manage their own profiles within the Weborder System. Their accounts must already be registered before they can log in. Once logged in, sales people should have access to order management tools but never administrative functions that only an Administrator would handle. Customer accounts also need to be registered for them to use the system, but customers should not have the ability to manage orders or other sensitive information. It's important that we keep these roles separate to ensure proper permissions and security.

[SYSTEM] Can you specify the exact permissions a Customer should have when they are added to the Weborder System, such as accessing order history or viewing product details?

[USER] Customers should be able to add items to their cart and view product details. When they log in, the system should allow them to access their order history but never let them modify or delete orders that aren't theirs. It's important because we want to protect user data and ensure only authorized users can make changes. We've had issues with unauthorized modifications before, so this is a big concern for us.

[SYSTEM] Could you specify the exact conditions under which the system should initiate a backup of critical data, such as during off-peak hours or before performing maintenance?

[USER] To ensure high availability and ease of migration, we need the system to initiate a backup of critical data during off-peak hours. This is important because it minimizes disruptions for our users. Additionally, before performing any maintenance tasks, such as software updates or server checks, the system should automatically back up all critical data. We aim for an availability rate of 99.99%, which means that we need to ensure minimal downtime and quick recovery in case of issues.

[SYSTEM] How often should the system automatically initiate a backup of critical data to ensure minimal data loss in case of hardware failure?

[USER] To ensure minimal data loss in case of hardware failure, we need the system to automatically initiate a backup of critical data at least once every hour. This way, if something goes wrong with our primary storage, we can quickly recover from the last backup without losing too much information. It should be easy for us to migrate or back up via another USB drive as per AUTO-0035.

[SYSTEM] Can you specify the exact hardware requirements and constraints for the system's deployment environment, such as network connectivity or power supply stability?

[USER] For our system to operate smoothly, we need a fixed hardware setup that can be updated via an internet connection. It's crucial for us to have reliable network connectivity so that we can patch and maintain the system as needed without downtime. We cannot afford any interruptions in this process.

Part C - Extracted requirements by category

Functional

- FR-001** It should be able to handle different types of products and quantities.
- FR-002** Yoggie should work closely with our organization on future enhancements and new features.
- FR-003** The system must exhibit specific functionalities under certain conditions, which is important for us because it ensures that all the necessary features are covered.
- FR-004** We need the system to send emails to customers and vendors as quickly as possible — ideally in less than a second.
- FR-005** For both customers and vendors, these emails should be triggered under specific conditions such as order confirmations, payment reminders, account updates, and support requests.
- FR-006** For our e-commerce platform, it should be able to process transactions quickly — ideally within less than 2 seconds from when a customer clicks 'buy now'.
- FR-007** We've had issues with slow login times, so we need the system to authenticate users efficiently, ensuring that it never takes more than 5 seconds for a user to log in.
- FR-008** For returning customers, the system should remember their card details but verify the card is still valid upon each login.
- FR-009** Additionally, the system must support multiple browsers including Chrome, Firefox, Safari, and Edge on both Windows and macOS operating systems.
- FR-010** The user interface should display product titles and metadata prominently but never show any private information like credit card numbers or passwords in the browser interface.
- FR-011** Only authorized users, identified by their unique login credentials, should have access to certain administrative features.
- FR-012** We must not allow unauthorized users to modify or view sensitive data.
- FR-013** Lastly, we need robust logging and auditing capabilities for all user activities, ensuring that every action is recorded and can be traced back to a specific user.
- FR-014** Sales people need to be able to place orders and manage their own profiles within the Weborder System.
- FR-015** Their accounts must already be registered before they can log in.
- FR-016** Once logged in, sales people should have access to order management tools but never administrative functions that only an Administrator would handle.
- FR-017** Customer accounts also need to be registered for them to use the system, but customers should not have the ability to manage orders or other sensitive information.
- FR-018** Customers should be able to add items to their cart and view product details.
- FR-019** When they log in, the system should allow them to access their order history but never let them modify or delete orders that aren't theirs.
- FR-020** It's important because we want to protect user data and ensure only authorized users can make changes.

Non-Functional

NFR-001	Additionally, the system should automatically detect and block IP addresses that show signs of DOS attacks.
NFR-002	if someone tries to log in multiple times without success, it should lock them out after a certain number of failed attempts.
NFR-003	During peak usage times, such as when we have a large number of concurrent users, the system should be able to handle searches efficiently.
NFR-004	For example, it should use Dijkstra's shortest path algorithm for searching, and it needs to perform these searches quickly — ideally in less than 1 second per search.
NFR-005	Specifically, it should be able to restore 1000 records per second and retrieve up to 200 products per second without any significant slowdowns.
NFR-006	Additionally, before performing any maintenance tasks, such as software updates or server checks, the system should automatically back up all critical data.
NFR-007	This way, if something goes wrong with our primary storage, we can quickly recover from the last backup without losing too much information.
NFR-008	It should be easy for us to migrate or back up via another USB drive as per AUTO-0035.

Data

DATA-001	This validation should occur every time a new email address is entered or an existing one is updated.
DATA-002	The system must use a SQL-based database for both inventory and other data storage to ensure consistency and reliability.

Business Rules

BR-001	Ideally it would happen automatically at least once a day, but we should also have the option to manually trigger these tests whenever necessary.
BR-002	For sure, we need the system to send out automated test emails during scheduled maintenance and after software updates.
BR-003	During maintenance windows, say between 8 PM and 2 AM on Tuesday nights, the system should automatically check if all outgoing emails are working properly.
BR-004	Similarly, right after a new version of the software is deployed, we want to make sure everything is still functioning as expected, so the system should send out test emails then too.
BR-005	This way, any issues can be caught early and addressed before they affect our users.
BR-006	Specifically, the system should start backing up critical data between 2 AM and 6 AM local time.
BR-007	Additionally, before performing maintenance tasks such as software updates or server checks, it should complete a backup to ensure that any changes are safe and can be reverted if necessary.

Interfaces

IF-001	For customer emails, we need the system to send an email confirmation that includes the items ordered and their tracking number.
IF-002	This should happen automatically after the order is placed.
IF-003	The web interface must be consistent across different browsers like Chrome, Firefox, and Safari, ensuring a user-friendly experience for our customers.
IF-004	For customer support queries, the system should send an email to our System Administrator whenever a technical issue is raised by customers or sales staff.
IF-005	The content of these emails should include details like the user's contact information and the nature of the problem so that it can be addressed promptly.

IF-006	This email should contain the tracking number for easy reference.
IF-007	Customers must access the web order system through a secure HTTPS connection from any compatible browser.
IF-008	We need the system to send an email confirmation to customers as soon as their order is placed.
IF-009	This email should include details about the items they ordered and the shipping address, along with a tracking number for easy reference.
IF-010	It's important because it helps ensure that customers are aware of their order status and can track their deliveries.

Constraints

C-001	For our system to operate smoothly, we need a fixed hardware setup that can be updated via an internet connection.
C-002	It's crucial for us to have reliable network connectivity so that we can patch and maintain the system as needed without downtime.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
Operating Environment Web Store shall operate with the following internet browsers: Microsoft Internet Explorer version 6 and 7, Netscape Communicator Version 4 and 5.	✗	— not recovered
Web Store shall operate on an Intel based system with Slackware Linux 2.6 and Apache Web Server.	✗	— not recovered
The system shall use SQL based database to store inventory information.	✓	The system must use a SQL-based database for both inventory and other data storage to ensure consistency and reliability.
Design and Implementation Constraints Must use a SQL based database.	✗	— not recovered
User Documentation Shall install online help for users via the web interface Shall deliver Operations and Maintenance manual, Users Guide book, and Installation Instructions for the Administrator.	✗	— not recovered
The interface must remain consistent among various web browsers and be intuitive to the customer.	✓	The web interface must be consistent across different browsers like Chrome, Firefox, and Safari, ensuring a user-friendly experience for our customers.
SI-1.3: Plug-ins interface Communications Interfaces CI-1: The WebOrder system shall send an e-mail confirmation to the customer that the items they ordered will be delivered to the shipping address along with tracking number.	✗	— not recovered
CI-2: The WebOrder system shall send an e-mail to System Administrator regarding any technical queries from customers or sales people.	✓	For customer support queries, the system should send an email to our System Administrator whenever a technical issue is raised by customers or sales staff.
Quality Attribute Requirements Upon the USB being plugged in the system shall be able to be deployed and operational in less than 1 minute.	✗	— not recovered
The system shall be able to handle 1000 customers logged in concurrently at the same time.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The system shall be able to retrieve 200 products per second.	✗	— not recovered
The system shall be able to add product to shopping cart in less than 2ms.	✓	For our e-commerce platform, it should be able to process transactions quickly — ideally within less than 2 seconds from when a customer clicks 'buy now'.
The system shall be able to search for a specified product in less than 1 second.	✓	During peak usage times, such as when we have a large number of concurrent users, the system should be able to handle searches efficiently.
The system shall be able to email customer and vendor in less than 1 second.	✓	We need the system to send emails to customers and vendors as quickly as possible — ideally in less than a second.
The system shall be able to validate credit card in less than 2 seconds.	✗	— not recovered
The system shall be able to acquire shipping charges in less than 2 seconds.	✗	— not recovered
The system shall be able to restore 1000 records per second.	✓	Specifically, it should be able to restore 1000 records per second and retrieve up to 200 products per second without any significant slowdowns.
Security Requirements The system shall validate credit cards against fraud.	✗	— not recovered
The system shall encrypt all sensitive information via https.	✗	— not recovered
The system shall encrypt all customer data in database.	✗	— not recovered
The system shall auto detect IP DOS attacks and block IP automatically.	✓	Additionally, the system should automatically detect and block IP addresses that show signs of DOS attacks.
The system shall detect consecutive failed login attempts.	✓	if someone tries to log in multiple times without success, it should lock them out after a certain number of failed attempts.
The system shall be protected by open source firewall called Firestarter.	✗	— not recovered
http://www.fs-security.com/ Availability Requirements The system shall have an availability of 99.99%.	✗	— not recovered
Efficiency Requirements The system shall perform searches via Dijkstra's shortest path algorithm.	✓	For example, it should use Dijkstra's shortest path algorithm for searching, and it needs to perform these searches quickly — ideally in less than 1 second per search.
For returning customers, the system shall validate 'existing' credit card in system after each log in.	✓	For returning customers, the system should remember their card details but verify the card is still valid upon each login.
The system shall automatically compress image files that are too large in size.	✗	— not recovered
The system shall validate email address existence.	✓	This validation should occur every time a new email address is entered or an existing one is updated.
The system shall utilize consistent symbols and colors for clear notifications.	✗	— not recovered
Maintainability Requirements The system shall utilize interchangeable plugins.	✗	— not recovered
The system shall be easily updatable for fixes and patches.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The system shall create logs of all changes, updates, or fixes that are done to the site.	✗	— not recovered
The system shall be easy to upgrade.	✗	— not recovered
Portability Requirements The system shall be extremely portable via the usb drive.	✗	— not recovered
The system shall be easy to migrate or backed up via another usb drive.	✓	It should be easy for us to migrate or back up via another USB drive as per AUTO-0035.
Testability Requirements The system should be able to run under debug mode.	✗	— not recovered
The system should be able to run test credit card transactions.	✗	— not recovered
The system should be able to run test shipping orders.	✗	— not recovered
The system should be able to create test environment of weborder system.	✗	— not recovered
Other Requirements The system hardware shall be fixed and patched via an internet connection.	✓	For our system to operate smoothly, we need a fixed hardware setup that can be updated via an internet connection.
Yoggie shall coordinate on future enhancement and features with our organization.	✓	Yoggie should work closely with our organization on future enhancements and new features.
Functional requirements: A statement of a piece of required functionality or a behavior that a system will exhibit under specific conditions.	✓	The system must exhibit specific functionalities under certain conditions, which is important for us because it ensures that all the necessary features are covered.
Nonfunctional requirements: A description of a property or characteristic that the system should exhibit.	✗	— not recovered
Pre condition: A condition that must be satisfied or a state the system must be in before a use case may begin.	✗	— not recovered
Requirement: A statement of a customer need or objective or of a condition or capability that a product must possess to satisfy such a need or objective.	✗	— not recovered
Actors: Customer Weborder System Preconditions: Customer must be able to access the web order system via a web browser with HTTPS.	✓	Customers must access the web order system through a secure HTTPS connection from any compatible browser.
Actors: Customer Weborder System Preconditions: Customer account must already be registered.	✓	Customer accounts also need to be registered for them to use the system, but customers should not have the ability to manage orders or other sensitive information.
Actors: Customer Weborder System Preconditions: Customer must be logged-in on the system.	✗	— not recovered
Add Item To Cart Goal Customer adds item(s) in the cart Actors: Customer Weborder System Preconditions: The customer must be logged-in on the system.	✗	— not recovered
Customer must have item(s) in the shopping cart.	✓	Customers should be able to add items to their cart and view product details.
Actors: Administrator Weborder System Preconditions: Administrator account must already be registered.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Actors: Administrator Weborder System Preconditions: Administrator must be logged-in on the system.	✓	Once logged in, sales people should have access to order management tools but never administrative functions that only an Administrator would handle.
Add User Goal Register a new customer, sales person, or administrator account with the system Actors: Administrator Weborder System Preconditions: Administrator must be able to access the weborder system via a web browser with HTTPS.	✗	— not recovered
Actors: Administrator Weborder System Preconditions: Administrator must be able to access the weborder system via a web browser with HTTPS.	✗	— not recovered
Actors: Sales Person Weborder System Preconditions: Sales Person account must already be registered.	✗	— not recovered
Actors: Sales Person Weborder System Preconditions: Sales Person must be logged-in on the system.	✓	Sales people need to be able to place orders and manage their own profiles within the Weborder System.
Actors: Sales Person Weborder System Preconditions: Sales Person must be able to access the weborder system via a web browser with HTTPS.	✗	— not recovered

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.1837 / 0.1579 / 0.1698. Weighted P/R/F1: 0.3878 / 0.3333 / 0.3585.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	Yoggie shall coordinate on future enhancement and features with our organization.	Yoggie should work closely with our organization on future enhancements and new features.	The generated requirement accurately reflects the source, with 'work closely' being a faithful interpretation of 'coordinate' and all other critical details preserved. The modality shift from 'shall' to 'should' is minor in this context.
full	The system shall auto detect IP DOS attacks and block IP automatically.	Additionally, the system should automatically detect and block IP addresses that show signs of DOS attacks.	The generated requirement faithfully covers the source requirement, preserving all critical details regarding automatic detection and blocking of IP DOS attacks.
full	The interface must remain consistent among various web browsers and be intuitive to the customer.	The web interface must be consistent across different browsers like Chrome, Firefox, and Safari, ensuring a user-friendly experience for our customers.	The generated requirement faithfully covers the source, rephrasing 'intuitive to the customer' as 'user-friendly experience' and providing examples for 'various web browsers' without altering the core need or adding material conditions.
full	The system shall be easy to migrate or backed up via another usb drive.	It should be easy for us to migrate or back up via another USB drive as per AUTO-0035.	The generated requirement faithfully covers the source, preserving all critical details regarding ease of migration/backup via USB. The 'as per AUTO-0035' is a self-reference and not a material addition.
full	For returning customers, the system shall validate 'existing' credit card in system after each log in.	For returning customers, the system should remember their card details but verify the card is still valid upon each login.	The generated requirement faithfully covers the source, preserving all critical details regarding returning customers, existing credit card validation, and the trigger (after each login).

Verdict	Source requirement	Generated requirement	Judge reason
full	Other Requirements The system hardware shall be fixed and patched via an internet connection.	For our system to operate smoothly, we need a fixed hardware setup that can be updated via an internet connection.	The generated requirement faithfully covers the source, stating the need for a fixed hardware setup that can be updated (patched) via an internet connection. The added justification does not alter the core requirement.
full	Actors: Customer Weborder System Preconditions: Customer must be able to access the web order system via a web browser with HTTPS.	Customers must access the web order system through a secure HTTPS connection from any compatible browser.	The generated text faithfully captures the precondition for customer access to the web order system via a secure browser connection.
full	CI-2: The WebOrder system shall send an e-mail to System Administrator regarding any technical queries from customers or sales people.	For customer support queries, the system should send an email to our System Administrator whenever a technical issue is raised by customers or sales staff.	The generated requirement faithfully covers all critical details of the source requirement, including the sender, recipient, trigger, and content of the email.
full	Actors: Customer Weborder System Preconditions: Customer account must already be registered.	Their accounts must already be registered before they can log in.	The generated requirement faithfully states that accounts must be registered, which is the core precondition from the source.
partial	The system shall be able to email customer and vendor in less than 1 second.	We need the system to send emails to customers and vendors as quickly as possible — ideally in less than a second.	The generated requirement weakens the strict time constraint from 'less than 1 second' to 'ideally in less than a second'.
partial	The system shall use SQL based database to store inventory information.	The system must use a SQL-based database for both inventory and other data storage to ensure consistency and reliability.	The generated requirement expands the scope of data stored in the SQL database beyond just inventory and adds a justification not in the source.
partial	The system shall be able to restore 1000 records per second.	Specifically, it should be able to restore 1000 records per second and retrieve up to 200 products per second without any significant slowdowns.	The generated requirement preserves the core detail of restoring 1000 records per second but adds a new, distinct requirement about retrieving products.
partial	The system shall detect consecutive failed login attempts.	if someone tries to log in multiple times without success, it should lock them out after a certain number of failed attempts.	The generated requirement covers the detection of failed login attempts but adds a material consequence (locking out the user) that is not present in the source requirement, which only specifies detection.
partial	Efficiency Requirements The system shall perform searches via Dijkstra's shortest path algorithm.	For example, it should use Dijkstra's shortest path algorithm for searching, and it needs to perform these searches quickly — ideally in less than 1 second per search.	The generated requirement correctly identifies Dijkstra's algorithm for searches but adds a new performance constraint (less than 1 second per search) not present in the source.
partial	The system shall validate email address existence.	This validation should occur every time a new email address is entered or an existing one is updated.	The generated requirement adds a material condition by specifying when the email validation should occur, which is an addition to the core need of simply validating email address existence.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
Operating Environment Web Store shall operate with the following internet browsers: Microsoft Internet Explorer version 6 and 7, Netscape Communicator Version 4 and 5.	The web interface must be consistent across different browsers like Chrome, Firefox, and Safari, ensuring a user-friendly experience for our customers.	The generated requirement changes the specific browsers and versions and shifts the focus from operating environment compatibility to interface consistency.
Web Store shall operate on an Intel based system with Slackware Linux 2.6 and Apache Web Server.	For our system to operate smoothly, we need a fixed hardware setup that can be updated via an internet connection.	The generated requirement is too general, losing specific details about the hardware, OS, and web server, and adds an update mechanism not present in the source.
Design and Implementation Constraints Must use a SQL based database.	The system must use a SQL-based database for both inventory and other data storage to ensure consistency and reliability.	The generated requirement adds scope (inventory and other data storage) and justification (consistency and reliability) not present in the source, which is a pure constraint.
The interface must remain consistent among various web browsers and be intuitive to the customer.	Additionally, the system must support multiple browsers including Chrome, Firefox, Safari, and Edge on both Windows and macOS operating systems.	The generated requirement focuses on supporting multiple browsers and operating systems, losing the critical details of interface consistency and intuitiveness mentioned in the source. It also adds operating system details.
SI-1.3: Plug-ins interface Communications Interfaces CI-1: The WebOrder system shall send an e-mail confirmation to the customer that the items they ordered will be delivered to the shipping address along with tracking number.	We need the system to send an email confirmation to customers as soon as their order is placed.	The generated requirement loses all critical details about the content of the email (items ordered, shipping address, tracking number) and adds a new condition about when the email should be sent.
SI-1.3: Plug-ins interface Communications Interfaces CI-1: The WebOrder system shall send an e-mail confirmation to the customer that the items they ordered will be delivered to the shipping address along with tracking number.	This email should include details about the items they ordered and the shipping address, along with a tracking number for easy reference.	The generated requirement loses the explicit 'confirmation' aspect and weakens the detail about the items 'being delivered to' the shipping address, instead just stating the address is included.
SI-1.3: Plug-ins interface Communications Interfaces CI-1: The WebOrder system shall send an e-mail confirmation to the customer that the items they ordered will be delivered to the shipping address along with tracking number.	This email should contain the tracking number for easy reference.	The generated requirement only preserves the detail about the tracking number, losing the critical details about the email being a confirmation to the customer and including information about items ordered and shipping address.
CI-2: The WebOrder system shall send an e-mail to System Administrator regarding any technical queries from customers or sales people.	This email should include details about the items they ordered and the shipping address, along with a tracking number for easy reference.	The generated requirement discusses email content, but the content (order details) and implied recipient are completely different from the source's email about technical queries to a System Administrator.
The system shall be able to handle 1000 customers logged in concurrently at the same time.	During peak usage times, such as when we have a large number of concurrent users, the system should be able to handle searches efficiently.	The generated requirement addresses system performance under concurrent users, similar to the source, but it loses the specific number of users (1000) and narrows the scope to handling 'searches efficiently' rather than the general ability to handle logged-in customers.
The system shall be able to retrieve 200 products per second.	Specifically, it should be able to restore 1000 records per second and retrieve up to 200 products per second without any significant slowdowns.	Adds a new performance requirement for restoring records and slightly weakens the original '200 products per second' with 'up to'.

Document: 1999_dii

Part A - Document header

Document ID	1999_dii
Generation model	ollama / qwen2.5:7b-instruct
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	22
Generated requirement count	14
Matched count	2
Clarification rounds	requested 2 / used 0
Clarification chunks	0
Semantic P / R / F1	0.1429 / 0.0909 / 0.1111
LLM strict P / R / F1	0.0000 / 0.0000 / 0.0000
LLM weighted P / R / F1	0.3214 / 0.2045 / 0.2500
Evidence / Propositions / Gap	64 / 14 / 0
Grounded after rewrite	14

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 1999 - dii system, and the goal is to capture the main requirements and constraints for document 1999_dii.

[SYSTEM] Could you provide more details on how the data management services will be adjusted by on-site personnel during varying workloads and database sizes?

[USER] Our data management services need to be flexible so that on-site personnel can adjust them based on varying workloads and database sizes. Specifically, we want various classes of users to have access: data consumers, data and database managers, and network information infrastructure resource managers. In our fixed mode of operation, it should be possible for the team to tune these services as needed. This is important because we need to ensure smooth performance regardless of how much data or traffic we're handling.

[SYSTEM] How will the data validation processes be adapted by on-site personnel to ensure they meet the T, D, A, and I qualification methods during different workload scenarios?

[USER] Our team needs to be able to apply the T, D, A, and I qualification methods during different workload scenarios. During testing (T), we need to ensure that our on-site personnel can run thorough checks under various conditions. For demonstrations (D), it should be clear how the system performs in real-time usage. Analysis (A) is important for deep diving into data and identifying any issues, especially when load increases. Inspection (I) will help us catch any potential problems before they become critical. We need to make sure these methods can be adapted flexibly depending on the workload — whether it's a small team or handling 1000 concurrent users. It should also work across different browsers and operating systems, like Chrome, Firefox, Windows, and macOS.

[SYSTEM] Can you explain how the on-site personnel will manage and adjust the data access services APIs as specified in Appendix C during different workload scenarios?

[USER] In our operations, we need the system to allow on-site personnel to manage and adjust data access services APIs as specified in Appendix C. Specifically, it should be able to redefine or reset names of connect descriptors for database server instances. The system must internally interface with existing data elements from DBIF, DAC, and COTS RDBMS products without requiring operator intervention. Additionally, the XS function needs to provide standard APIs as outlined in Appendix C. This is all part of our broader software capabilities required for the DII COE components.

[SYSTEM] How will the compliance level requirements described in the I&RTS be enforced and monitored by on-site personnel during different workload scenarios?

[USER] Compliance level requirements described in the I&RTS will be enforced and monitored by on-site personnel through adherence to these standards. COE sponsors are responsible for ensuring that all operations comply with these requirements. For the deployment environment, developers of the system embedding the XS module need to determine specific training needs to ensure everyone is up to speed on how to meet these compliance levels during different workload scenarios.

[SYSTEM] How will the session time-out values be configured and managed by on-site personnel to ensure they align with varying user roles and workloads?

[USER] We need to be able to configure session time-out values based on user roles and workloads. For example, super-users might have longer sessions than regular users. It should allow us to set these values before a user logs in and also let us adjust them as needed without affecting the current active sessions. Maintenance tasks like shutting down or upgrading the system should not disrupt this setting process. We want to ensure that all modifications and upgrades are handled smoothly, including when we go online or offline.

[SYSTEM] How will the Registry software's portability and execution on COE-compliant platforms impact the deployment process and potential challenges for on-site personnel during different workload scenarios?

[USER] Portability and execution on COE-compliant platforms are crucial for us. The Registry software needs to be flexible enough to run smoothly across different environments, which means it should work well on any COE-compliant platform we choose. This is important because our deployment process will need to accommodate various scenarios, from small-scale testing to large-scale production with up to 1000 concurrent users. On-site personnel might face challenges during these deployments, especially when dealing with different workload conditions. We want the software to handle everything from light usage periods to peak times without any issues. Ideally, it should be able to scale efficiently and maintain performance even under heavy load. Security is also a big concern; we need to ensure that all data remains encrypted both in transit and at rest. That's important to us because we cannot afford any breaches. We want the system to validate credit card information for returning users but not require them to re-enter everything every time they log in.

[SYSTEM] How will the maintainability of the Registry software be ensured during different workload scenarios, particularly in terms of ease of updating and debugging?

[USER] Maintaining the Registry software during different workload scenarios is crucial for us. We need a system that allows easy updates and debugging to ensure smooth operation under various conditions. For instance, when we have more than 1000 concurrent users, it should handle the load without any issues. The software should be designed with clear architecture and modular components so that updating or fixing bugs becomes straightforward. Ideally, it would have a robust logging system in place for easy debugging. We've had issues before where updates caused unexpected downtime, so we need to ensure that changes can be made without disrupting service. That's important to us because our users expect consistent performance and reliability.

[SYSTEM] How will the testing process for the Registry software be structured to ensure it meets maintainability and portability requirements during different workload scenarios?

[USER] To ensure the Registry software meets maintainability and portability requirements during different workload scenarios, we need a comprehensive testing process. This should include automated tests for both functional and non-functional aspects of the system. We'll use load testing to simulate up to 1000 concurrent users and stress testing to push the limits beyond that. It's crucial that our tests cover various environments — including different browsers like Chrome, Firefox, and Safari on Windows, macOS, and Linux. For security, we must verify that all customer data is encrypted both in transit and at rest. We also need to ensure that the system can be easily ported to other platforms if needed, so testing should include cross-platform compatibility checks. Additionally, it's important that our tests validate the system's ability to handle different types of user roles and permissions without conflicts. Ideally, we want a robust logging mechanism in place for debugging and auditing purposes. That's important to us because maintaining a clear trail is key for compliance and troubleshooting.

[SYSTEM] How will the Registry software's maintainability be ensured during updates and debugging processes, especially when transitioning between different COE-compliant platforms?

[USER] Maintaining the Registry software's portability and testability is crucial for us. We need to ensure that when we update or debug the system, especially during transitions between different COE-compliant platforms, it should be easy to switch without causing disruptions. Ideally, it would have a consistent codebase that can adapt to various environments. The problem we have is ensuring that all components are well-documented and modular so they can be easily replaced or updated. We've had issues with legacy systems where changes in one area caused unexpected problems elsewhere. That's important to us because maintaining the system should not require extensive rework every time we make a change. It should be able to run on multiple platforms without needing significant modifications, and it must support automated testing across all environments to catch any issues early.

[SYSTEM] Could you provide more details on the specific metadata fields and formats that need to be supported for contributing to the registry?

[USER] For contributing to the registry, we need our system to be able to receive XML registry forms. The producer should be able to submit metadata and related information resource artifacts. It's important that this process is seamless and efficient. We also need the system to render semantically valid views from an XML document, like XSL transformations, which will disassemble hierarchical views into a relational representation. This needs to align with standard APIs as specified in Appendix B of our documentation. The goal here is to ensure that all metadata fields are correctly formatted and can be easily integrated into the registry services.

Part C - Extracted requirements by category

Functional

FR-001 The software should allow easy updates and debugging for smooth operation under various conditions.

Non-Functional

- NFR-001** The system should handle up to 1000 concurrent users without issues.
- NFR-002** The software must support automated testing across all environments to catch issues early.
- NFR-003** The system should be designed with clear architecture and modular components.
- NFR-004** The software should be able to run on multiple platforms without significant modifications.
- NFR-005** The system should validate credit card information for returning users without requiring re-entry.
- NFR-006** The software should handle different types of user roles and permissions without conflicts.
- NFR-007** The system should be able to scale efficiently and maintain performance under heavy load.
- NFR-008** The software must ensure data encryption both in transit and at rest.
- NFR-009** The system should be easily ported to other platforms if needed.
- NFR-010** The software should handle different environments, including various browsers.
- NFR-011** The system should be able to receive and process XML registry forms.
- NFR-012** The software should render semantically valid views from XML documents, like XSL transformations.
- NFR-013** The system should be designed with a robust logging mechanism for debugging and auditing purposes.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
DOCUMENT OVERVIEW This document outlines the software capabilities required for the XS components for the DII COE.	X	— not recovered
COE sponsors shall adhere to compliance level requirements described in the I&RTS.	X	— not recovered
In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	X	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The XS shall have the ability to redefine or reset names of connect descriptors to database server instances.	X	— not recovered
At a minimum, the session time-out values shall be user definable and be able to be reset prior to initialization of a user session.	X	— not recovered
End User Mode: Portion of XS services shall be used by various classes of users: data consumers, data and database managers, network information infrastructure resource managers.	X	— not recovered
Some of these uses of the data management services will entail unique requirements that shall be fulfilled within the capability of XS services.	X	— not recovered
In addition, maintenance also shall pertain to shutdown, open not-mounted and online/off-line implementations, modifications, upgrade, or other related actions.	X	— not recovered
The data management services shall support managing various types of data, database architectures and platforms that includes hardware and software at the specified sites.	X	— not recovered
In support of training activities, the data management services shall provide for the same processing as would be encountered in a production environment.	X	— not recovered
To contribute metadata to a registry, the producer must be able to receive XML registry forms, submit metadata and the related information resource artifacts, and notify.	X	— not recovered
Extract semantically-valid views from an XML document (e.g., XSL) Disassemble hierarchical view to relational representation Render REGISTRY SERVICES The Registry shall provide standard APIs as specified in Appendix B.	✓	The software should render semantically valid views from XML documents, like XSL transformations.
DATA ACCESS SERVICES INTERNAL INTERFACE REQUIREMENTS The XS function shall provide standard APIs as specified in Appendix C.	X	— not recovered
The XS shall internally interface (transparent to the operator) with the existing data elements of the DBIF, DAC, and COTS RDBMS products.	X	— not recovered
REGISTRY SERVICES ENVIRONMENT REQUIREMENTS The Registry software shall be portable and required to execute on COE-compliant platforms.	✓	The system should be able to receive and process XML registry forms.
Personnel requirements shall be determined by the developers of the system in which the XS module is embedded TRAINING-RELATED REQUIREMENTS Not applicable.	X	— not recovered
Training requirements shall be determined by the developers of the system in which the XS module is embedded.	X	— not recovered
OTHER REQUIREMENTS None PACKAGING REQUIREMENTS The XS software shall be delivered in accordance with DII COE guidelines.	X	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The Qualification Methods applied to the software shall include test, demonstration, analysis, and inspection (T, D, A, I).	X	— not recovered
Facts are either links (relationships) between entities or associations between an entity and attributes Constraints are conditions about the data that must always be true.	X	— not recovered
If, after reviewing the tags in the Registry, you cannot reuse an existing specification (a.k.a.	X	— not recovered
But if they were identified as attributes of the element ctry, then the additional baggage (other attributes) must be carried, or submitted as separate elements.	X	— not recovered

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.0000 / 0.0000 / 0.0000. Weighted P/R/F1: 0.3214 / 0.2045 / 0.2500.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
partial	Extract semantically-valid views from an XML document (e.g., XSL) Disassemble hierarchical view to relational representation Render REGISTRY SERVICES The Registry shall provide standard APIs as specified in Appendix B.	The software should render semantically valid views from XML documents, like XSL transformations.	The generated requirement only covers the 'render semantically valid views from XML documents' aspect of the source, omitting other critical details such as 'Extract semantically-valid views', 'Disassemble hierarchical view to relational representation', and the provision of 'standard APIs' by Registry Services.
partial	To contribute metadata to a registry, the producer must be able to receive XML registry forms, submit metadata and the related information resource artifacts, and notify.	The system should be able to receive and process XML registry forms.	The generated requirement only covers the ability to 'receive XML registry forms' and adds 'process', while omitting other critical details from the source such as 'submit metadata and the related information resource artifacts, and notify' and the overall goal 'To contribute metadata to a registry'.
partial	In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	The system should be able to scale efficiently and maintain performance under heavy load.	The generated requirement is a generalization. It captures the idea of handling varying loads but loses critical details like "tuned by on-site personnel", "data management services", "fixed mode", and "adjusting for database sizes".
partial	In addition, maintenance also shall pertain to shutdown, open not-mounted and online/off-line implementations, modifications, upgrade, or other related actions.	The software should allow easy updates and debugging for smooth operation under various conditions.	The generated requirement covers 'updates' (modifications/upgrade) but omits other specific maintenance actions like shutdown, open not-mounted, and online/off-line implementations. It also adds 'debugging' which is not explicitly in the source.

Verdict	Source requirement	Generated requirement	Judge reason
partial	REGISTRY SERVICES ENVIRONMENT REQUIREMENTS The Registry software shall be portable and required to execute on COE-compliant platforms.	The software should be able to run on multiple platforms without significant modifications.	The generated requirement captures portability but misses the critical detail of 'COE-compliant platforms' and weakens the modality from 'shall' to 'should'.
partial	The Qualification Methods applied to the software shall include test, demonstration, analysis, and inspection (T, D, A, I).	The software must support automated testing across all environments to catch issues early.	The generated requirement only covers 'test' from the source and adds the detail of 'automated testing across all environments' and a purpose, while omitting 'demonstration, analysis, and inspection'.
partial	End User Mode: Portion of XS services shall be used by various classes of users: data consumers, data and database managers, network information infrastructure resource managers.	The software should handle different types of user roles and permissions without conflicts.	The generated requirement correctly identifies the need for handling different user types but generalizes the specific user classes mentioned in the source and adds a condition ('without conflicts').
partial	COE sponsors shall adhere to compliance level requirements described in the I&RTS.	The software must ensure data encryption both in transit and at rest.	The generated requirement introduces a specific technical solution (data encryption) and applies it to software, whereas the source is a general compliance adherence requirement for COE sponsors referencing an external document. It loses critical details about the actor and the source of compliance.
partial	The data management services shall support managing various types of data, database architectures and platforms that includes hardware and software at the specified sites.	The software should handle different environments, including various browsers.	The generated requirement narrows the scope of 'various types of data, database architectures and platforms' to 'different environments, including various browsers,' losing critical details about data, database architectures, platforms, hardware, software, and sites. While 'environments' is related, 'browsers' is a specific and limited interpretation.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	The system should handle up to 1000 concurrent users without issues.	The generated requirement introduces a specific number of concurrent users, which is a material addition and a different metric than "varying workloads and sizes of associated databases". It also loses all other critical details from the source.
Extract semantically-valid views from an XML document (e.g., XSL) Disassemble hierarchical view to relational representation Render REGISTRY SERVICES The Registry shall provide standard APIs as specified in Appendix B.	The system should be able to receive and process XML registry forms.	The generated requirement covers XML processing but misses the API provision and specific view extraction/disassembly details from the source, and adds 'registry forms'.
REGISTRY SERVICES ENVIRONMENT REQUIREMENTS The Registry software shall be portable and required to execute on COE-compliant platforms.	The system should be easily ported to other platforms if needed.	The generated requirement captures the portability aspect but weakens the modality from "shall be required" to "should be easily ported" and generalizes "COE-compliant platforms" to "other platforms", losing a specific constraint.

Source requirement	Generated requirement	Judge reason
REGISTRY SERVICES ENVIRONMENT REQUIREMENTS The Registry software shall be portable and required to execute on COE-compliant platforms.	The software should handle different environments, including various browsers.	The generated requirement broadly mentions handling different environments, which relates to portability, but it loses the specific "COE-compliant platforms" constraint and adds "including various browsers" which is a specific, potentially unrelated, detail.
If, after reviewing the tags in the Registry, you cannot reuse an existing specification (a.k.a.	The system should be able to receive and process XML registry forms.	The generated requirement mentions 'registry' but shifts the focus from reviewing tags and reusing specifications to processing 'XML registry forms', losing critical details and adding new material.
But if they were identified as attributes of the element ctry, then the additional baggage (other attributes) must be carried, or submitted as separate elements.	The software should render semantically valid views from XML documents, like XSL transformations.	While both relate to XML, the generated requirement focuses on rendering views from XML documents, which is a different concern from the source's specific constraint on how attributes of an element must be carried or submitted. It loses the critical detail of the source and introduces a new material aspect.

Document: 2005_microcare

Part A - Document header

Document ID	2005_microcare
Generation model	ollama / qwen2.5:7b-instruct
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	25
Generated requirement count	19
Matched count	12
Clarification rounds	requested 2 / used 2
Clarification chunks	2
Semantic P / R / F1	0.6316 / 0.4800 / 0.5455
LLM strict P / R / F1	0.1579 / 0.1200 / 0.1364
LLM weighted P / R / F1	0.3684 / 0.2800 / 0.3182
Evidence / Propositions / Gap	47 / 19 / 0
Grounded after rewrite	19

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 2005 - microcare system, and the goal is to capture the main requirements and constraints for document 2005_microcare.

[SYSTEM] Could you provide more details on the specific information that the Salesman master needs to capture during a sale?

[USER] For a sale, the Salesman master needs to capture several key pieces of information. Firstly, it should be able to record the customer's name and contact details. Secondly, we need to track the product or service being sold, including its ID and description. The quantity sold is also crucial. Additionally, the price per unit and total sale amount must be captured accurately. We want to ensure that payment method information is recorded for each transaction. It should also log the date and time of the sale, along with any special notes or comments from the Salesman. This will help us keep track of sales trends and customer interactions.

[SYSTEM] Can you specify which details need to be captured during the claim entry process and how they might differ from the sales information?

[USER] During the claim entry process, we need to capture detailed information such as the claimant's name, contact details, date of birth, and a description of the incident. This is important because it helps us verify the identity and ensure accurate claims processing. The system should also log the IP address and timestamp for each entry to maintain audit trails. Regarding sales information, we only need to capture basic transaction data like order ID and total amount, but not any personal details. Authorized users can amend or alter claim entries if needed, which is crucial for correcting errors or updating claims as necessary.

[SYSTEM] Could you elaborate on how the system should handle duplicate staff entries based on the VSP Name and Staff Code?

[USER] When adding a new staff member, the system should check for existing entries based on VSP Name and Staff Code. If there's a duplicate, it should generate a message alerting the user. However, if the user needs to save this new entry despite the duplication, they should be allowed to do so. This is important because we have had issues with staff members having similar names or codes in our system. The principles for handling such cases are defined in our PDS document, ensuring that only valid and necessary entries are saved. Additionally, it's crucial that each voucher slip contains a barcode with two unique identifiers: one for the client and another for the partner.

[SYSTEM] Could you clarify how the system should handle voucher selection and validation for different levels of facilities based on the VSP guidelines?

[USER] For voucher selection and validation, we need to ensure that the system pulls data from their own master lists for Health Sub-District, Locality, Level Of Facility, Type of facility, and Registration Body based on VSP guidelines. The principles of the vouchers and health care services should be defined in our PDS, including quality standards and monitoring measures. When it comes to invoice amounts, we need to validate that they are calculated correctly as the number of vouchers multiplied by the wholesale price. For payments, cash is the only mode accepted during distribution transactions. If a claim (treatment form) status needs to be rejected, users should mark it as such and send it back to VSP for review.

[SYSTEM] Could you provide more details on how the system should handle the allocation of vouchers based on the number required by the distributor, including any specific criteria or rules?

[USER] For voucher allocation, the system should be able to allocate relevant vouchers based on the number required by the distributor. Specifically, it needs to match the number of forms entered during claim submission with the available forms in stock. The system should print a receipt document accordingly. Additionally, when allocating vouchers, the system must consider the distributor's name and the sales man involved, along with the date of distribution. Each voucher allocated should include its number and batch number. If the status is 'Rejection' or 'Quarantine', we need to capture the reason for rejection from a predefined master list. This ensures that all necessary details are recorded accurately.

[SYSTEM] Could you provide more details on how the system should populate the reason for referral when the patient status is selected as 'Referral', and ensure it comes from the Referral Master?

[USER] When the patient status is selected as 'Referral', we need to have a way for the system to capture both 'Referred To' and 'Reason for Referral'. The reason for referral should be populated automatically from the Referral Master. This ensures consistency and accuracy in our records. Additionally, it's important that if the claim status is either 'Rejection' or 'Quarantine', we have a provision to capture the reason for the rejection or quarantine. This will help us understand the issues better and improve our processes.

[SYSTEM] Can you provide more details on the process for adding new salesmen to the Team Master and how it integrates with user roles permissions?

[USER] For adding new salesmen to the Team Master, only system admins should be able to do this. They can create any number of individual users under existing user groups. When a distribution transaction happens, the user needs to select the name of the salesman from the Team Master list. This ensures that every sale is linked to the correct team member.

[SYSTEM] Could you specify what information should be included in the document related to payment terms between MSIU and VSP?

[USER] We need the document to include all the details of the payment terms agreed between MSIU and VSP. It should capture everything that is discussed on the VSP Master Information screen, where we enter these terms. This is important because it ensures clarity and prevents any misunderstandings in the future.

[SYSTEM] Could you clarify the criteria for determining when a warning message and inactivation of an VSP should occur based on Thump Impression mismatches?

[USER] If a VSP has more than two instances of Thump Impression mismatches for the same claim, we need the system to automatically generate a warning message. At the same time, it should In-Activate that VSP. This is important because it helps us catch and address potential issues promptly without disrupting service for other users.

Part C - Extracted requirements by category

Functional

- FR-001** The Salesman master should record customer's name and contact details during a sale.
- FR-002** The Salesman master should track the product or service being sold, including its ID and description during a sale.
- FR-003** The Salesman master should record the quantity sold during a sale.
- FR-004** The Salesman master should capture the price per unit and total sale amount during a sale.
- FR-005** The Salesman master should record payment method information for each transaction.

- FR-006** The Salesman master should log the date and time of the sale, along with any special notes or comments from the Salesman.
- FR-007** During the claim entry process, we need to capture detailed information such as the claimant's name, contact details, date of birth, and a description of the incident. This is important because it helps us verify the identity and ensure accurate claims processing.
- FR-008** The system should log IP address and timestamp for each entry to maintain audit trails. Regarding sales information, we only need to capture basic transaction data like order ID and total amount, but not any personal details.
- FR-009** Authorized users should be able to amend or alter claim entries if needed.
- FR-010** The system should allocate relevant vouchers based on the number required by the distributor.
- FR-011** The system should match the number of forms entered during claim submission with available forms in stock.
- FR-012** The system should print a receipt document when allocating vouchers.
- FR-013** When allocating vouchers, the system must consider the distributor's name and the sales man involved, along with the date of distribution. Each voucher allocated should include its number and batch number.
- FR-014** The system should include the number and batch number of each voucher allocated.
- FR-015** System admins should be able to add new salesmen to the Team Master.

Data

- DATA-001** Each voucher slip should contain a barcode with two unique identifiers: one for the client and another for the partner.
- DATA-002** The system should validate invoice amounts as the number of vouchers multiplied by the wholesale price.
- DATA-003** Cash is the only mode accepted during distribution transactions.

Business Rules

- BR-001** If the claim status is either 'Rejection' or 'Quarantine', we have a provision to capture the reason for the rejection or quarantine. This will help us understand the issues better and improve our processes.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
The principles of the voucher, the contents of the health care services, their required quality, the standards to be met by the providers and their infrastructure, as well as the measures for monitoring the quality have been defined in the PDS.	✗	— not recovered
Each voucher should have two portions with three tear off voucher slips each for Client and Partner.	✓	The system should print a receipt document when allocating vouchers.
Each voucher should have the following properties, which will have sub-elements to get the batch numbers, voucher numbers, and the project codes.	✓	The system should include the number and batch number of each voucher allocated.
This validity date can be amended or altered at the system level ONLY by the authorized user Voucher will also have MRP (maximum retail price) Voucher should have three tear off portioned slips with a sub-section tear-off for the partner.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Each voucher slip should contain the bar code of the Voucher with two identifications one for the client and the other for the partner.	✓	Each voucher slip should contain a barcode with two unique identifiers: one for the client and another for the partner.
During every distribution transaction the user should select the name of the sales man listed from Team Master.	✓	System admins should be able to add new salesmen to the Team Master.
The Salesman master should capture following information's.	✓	The Salesman master should record customer's name and contact details during a sale.
Name of the distributor* Name of the Sales Man* Date of distribution* No of vouchers sold* According to the number of vouchers required by the distributor, the system should allocate the relevant vouchers with their numbers and batch numbers based on the stock.	✓	When allocating vouchers, the system must consider the distributor's name and the sales man involved, along with the date of distribution. Each voucher allocated should include its number and batch number.
Invoice amount = No of vouchers x Wholesale price Mode of payment is Cash The mandatory information required during a distribution transaction is mentioned below.	✓	The system should validate invoice amounts as the number of vouchers multiplied by the wholesale price.
While entering the No of vouchers required, the system should automatically pick the Batch No's with voucher No's from the available voucher stock and list the details of each voucher with below information's as a grid format.	✓	The system should allocate relevant vouchers based on the number required by the distributor.
The values of Health Sub – District, Locality, Level Of Facility, Type of facility, Registration Body would be list from their own master and should select the details based on the VSP.	✗	— not recovered
The VSP Master Information screen should have a provision to enter the payment terms agreed between MSIU and VSP.	✗	— not recovered
VSP Name Staff Code Staff Name Staff Type Qualification The system will generate message while creating a new staff with existing name, but the system will allow the user to save that new staff if its required.	✗	— not recovered
Claim (Treatment Form) Submission Depends on the payment terms (Bi-monthly / Monthly) mentioned in the VSP master the VSP should submit the Claim (Treatment form) to the MISU Field office.	✗	— not recovered
These information is vital and shall be used for compared with the processed claims.	✗	— not recovered
In the transaction part no of forms should match with No of Available Forms The system should print a receipt document based on the information entered during claim submission.	✓	The system should match the number of forms entered during claim submission with available forms in stock.
The document should have the following information.	✗	— not recovered
Before the claim entry the user should check the form manually to understand whether any mandatory information is missed in the Claim or not.	✗	— not recovered
If yes, then the user should mark that Claim (Treatment Form) status as Rejected and send back to VSP.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
During the claim entry the system should capture the following information.	✓	During the claim entry process, we need to capture detailed information such as the claimant's name, contact details, date of birth, and a description of the incident. This is important because it helps us verify the identity and ensure accurate claims processing.
Patient Status - (Select from list of status options populating from Patient Status Master) suppose the status is Referral then the system will have the provision to capture Referred To and Reason for Referral, the reason for Referral should be populated from Referral master.	✗	— not recovered
The system should have a provision to capture the reason suppose if the claim status is "Rejection or Quarantine" Claim Amount – The calculation of claim amount is explained below.	✓	If the claim status is either 'Rejection' or 'Quarantine', we have a provision to capture the reason for the rejection or quarantine. This will help us understand the issues better and improve our processes.
If the claim from same VSP having more than two times of Thump Impression mismatching, then the system should automatically produce a warning message and the same time the system should In-Activate the VSP.	✗	— not recovered
The authorized user can alter / amend the Claim if required.	✓	Authorized users should be able to amend or alter claim entries if needed.
The system admin only can able to create User Groups The system should have the provision to create any number of individual user under any user group.	✗	— not recovered

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.1579 / 0.1200 / 0.1364. Weighted P/R/F1: 0.3684 / 0.2800 / 0.3182.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	Each voucher slip should contain the bar code of the Voucher with two identifications one for the client and the other for the partner.	Each voucher slip should contain a barcode with two unique identifiers: one for the client and another for the partner.	The generated requirement faithfully covers the source, preserving all critical details about the barcode and its two identifiers for client and partner.
full	The authorized user can alter / amend the Claim if required.	Authorized users should be able to amend or alter claim entries if needed.	The generated requirement faithfully covers the source, preserving the core need for authorized users to amend claims.
full	Invoice amount = No of vouchers x Wholesale price Mode of payment is Cash The mandatory information required during a distribution transaction is mentioned below.	The system should validate invoice amounts as the number of vouchers multiplied by the wholesale price.	The generated requirement faithfully captures the calculation for the invoice amount as stated in the source, interpreting it as a validation rule.
partial	Name of the distributor* Name of the Sales Man* Date of distribution* No of vouchers sold* According to the number of vouchers required by the distributor, the system should allocate the relevant vouchers with their numbers and batch numbers based on the stock.	When allocating vouchers, the system must consider the distributor's name and the sales man involved, along with the date of distribution. Each voucher allocated should include its number and batch number.	The generated requirement loses critical details regarding the number of vouchers sold/required by the distributor and the condition of allocation being "based on the stock". It also weakens the modality for some aspects from "should allocate" to "must consider".

Verdict	Source requirement	Generated requirement	Judge reason
partial	Each voucher should have the following properties, which will have sub-elements to get the batch numbers, voucher numbers, and the project codes.	The system should include the number and batch number of each voucher allocated.	The generated requirement covers voucher numbers and batch numbers but omits the critical detail of 'project codes' and 'sub-elements' mentioned in the source.
partial	The system should have a provision to capture the reason suppose if the claim status is "Rejection or Quarantine" Claim Amount – The calculation of claim amount is explained below.	If the claim status is either 'Rejection' or 'Quarantine', we have a provision to capture the reason for the rejection or quarantine. This will help us understand the issues better and improve our processes.	The generated requirement fully covers the part of the source about capturing the reason for claim rejection or quarantine, preserving all critical details. However, it omits the second part of the source regarding 'Claim Amount' calculation.
partial	During the claim entry the system should capture the following information.	During the claim entry process, we need to capture detailed information such as the claimant's name, contact details, date of birth, and a description of the incident. This is important because it helps us verify the identity and ensure accurate claims processing.	The generated requirement correctly identifies the core need of capturing information during claim entry, but it materially adds specific examples of information to be captured and a justification, which were not present in the source text. The source only stated "the following information" without specifying what.
partial	The Salesman master should capture following information's.	The Salesman master should record customer's name and contact details during a sale.	The generated requirement specifies particular information to be captured, which is an addition to the very general source requirement that only states "capture following information's" without specifying what.
partial	In the transaction part no of forms should match with No of Available Forms The system should print a receipt document based on the information entered during claim submission.	The system should match the number of forms entered during claim submission with available forms in stock.	The generated requirement covers the matching of forms but completely misses the requirement to print a receipt document.
partial	While entering the No of vouchers required, the system should automatically pick the Batch No's with voucher No's from the available voucher stock and list the details of each voucher with below information's as a grid format.	The system should allocate relevant vouchers based on the number required by the distributor.	The generated requirement covers the core need of allocating vouchers based on quantity but loses critical details about automatically picking from stock, including batch/voucher numbers, and displaying details in a grid format.
partial	The system admin only can able to create User Groups The system should have the provision to create any number of individual user under any user group.	System admins should be able to add new salesmen to the Team Master.	The generated requirement generalizes 'User Groups' and 'individual user' to 'Team Master' and 'salesmen', losing specific details about user management and adding new specific details.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
Each voucher should have the following properties, which will have sub-elements to get the batch numbers, voucher numbers, and the project codes.	When allocating vouchers, the system must consider the distributor's name and the sales man involved, along with the date of distribution. Each voucher allocated should include its number and batch number.	The generated requirement adds significant material details about considering distributor's name, salesman, and date of distribution, which are not present in the source. It also omits 'project codes' and 'sub-elements'.

Source requirement	Generated requirement	Judge reason
The Salesman master should capture following information's.	The Salesman master should record payment method information for each transaction.	The generated requirement specifies particular information to be captured, which is an addition to the very general source requirement that only states "capture following information's" without specifying what.
The Salesman master should capture following information's.	The Salesman master should record the quantity sold during a sale.	The generated requirement specifies particular information to be captured, which is an addition to the very general source requirement that only states "capture following information's" without specifying what.
The Salesman master should capture following information's.	The Salesman master should track the product or service being sold, including its ID and description during a sale.	The generated requirement specifies particular information to be captured, which is an addition to the very general source requirement that only states "capture following information's" without specifying what.
The Salesman master should capture following information's.	The Salesman master should capture the price per unit and total sale amount during a sale.	The generated requirement specifies particular information to be captured, which is an addition to the very general source requirement that only states "capture following information's" without specifying what.
The Salesman master should capture following information's.	The Salesman master should log the date and time of the sale, along with any special notes or comments from the Salesman.	The generated requirement specifies particular information to be captured, which is an addition to the very general source requirement that only states "capture following information's" without specifying what.
Name of the distributor* Name of the Sales Man* Date of distribution* No of vouchers sold* According to the number of vouchers required by the distributor, the system should allocate the relevant vouchers with their numbers and batch numbers based on the stock.	The system should allocate relevant vouchers based on the number required by the distributor.	The generated requirement loses critical details about including voucher numbers and batch numbers, and that allocation should be based on stock.
While entering the No of vouchers required, the system should automatically pick the Batch No's with voucher No's from the available voucher stock and list the details of each voucher with below information's as a grid format.	The system should include the number and batch number of each voucher allocated.	The generated requirement loses critical details about automatically picking vouchers from stock and listing details in a grid format.
While entering the No of vouchers required, the system should automatically pick the Batch No's with voucher No's from the available voucher stock and list the details of each voucher with below information's as a grid format.	When allocating vouchers, the system must consider the distributor's name and the sales man involved, along with the date of distribution. Each voucher allocated should include its number and batch number.	The generated requirement adds material conditions (distributor's name, salesman, date of distribution) not present in the source requirement and loses critical details about automatically picking vouchers from stock and listing them in a grid format.
VSP Name Staff Code Staff Name Staff Type Qualification The system will generate message while creating a new staff with existing name, but the system will allow the user to save that new staff if its required.	System admins should be able to add new salesmen to the Team Master.	The generated requirement covers adding new staff but loses critical details about specific staff attributes (VSP Name, Staff Code, Staff Type, Qualification) and the system's behavior when creating a new staff with an existing name.